Motion to Meaning: Word-Level Bangla Sign Language Recognition Using Mid-Fusion of CNN and TCN

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Sign Language Recognition (SLR) has seen sustained progress, yet many systems remain template-centric or unimodal and degrade under real-world variation in lighting, background, and signer style. These challenges are more severe for Bangla Sign Language (BdSL), due to limited word-level datasets. We present a scalable BdSL word-level recognition framework that uses mid-fusion to combine an RGB video stream (R3D-18) with a landmark stream (SE-augmented Temporal Convolutional Network), enabling robust spatio-temporal feature learning. The pipeline includes efficient preprocessing (frame sampling, adaptive ROI cropping, motion-aware selection, landmark normalization, and background suppression) to improve reliability in unconstrained settings. We also introduce BdSL-300, an open-source dataset of 300 words with roughly 8,000 isolated-sign videos from 20 signers. Our developed model achieves 97.4% Top-1 and 99.99% Top-5 test accuracy with strong category-level performance (macro F1 $\approx$ 0.9269), indicating that intermediate multimodal fusion is an effective and practical direction for accessible BdSL recognition.

Keywords: Sign Language Recognition, SLR, SLT, CNN, Kinetics-400, TCN, Edge Computing, HCI, Green AI

Table of Contents

Declaration **i**

Approval **ii**

Abstract **iii**

Table of Contents **iv**

List of Figures **vii**

List of Tables **viii**

Nomenclature **viii**

1 Introduction **1**
1.1 Background 1
1.2 Rational of the Study or Motivation 2
1.3 Problem Statement 2
1.4 Objective 3
1.5 Methodology in Brief 3
1.6 Scopes and Challenges 6

2 Literature Review **7**
2.1 Preliminaries 7
2.2 Review of Existing Research 9
2.3 Summary of Key Findings 15

3 Requirements, Impacts and Constraints **17**
3.1 Final Specifications and Requirements 17
3.1.1 System Specifications 17
3.1.2 Data Collection and Curation Requirements 17
3.1.3 Technical and Computational Requirements 18
3.2 Societal Impact 18
3.3 Environmental Impact 18
3.4 Ethical Issues 19
3.4.1 Data Ethics and Privacy 19
3.4.2 Algorithmic Fairness and Performance 20
3.5 Standards 20
3.6 Project Management Plan 20

4 Proposed Methodology **21**
4.1 Design Process or Methodology Overview 21
4.2 Model Benchmarking and Selection for System Integration 22
4.2.1 RGB-Based Model Evaluation 22
4.2.2 Landmark-Based Model Evaluation 24
4.3 Data Collection 25
4.4 Data Pre Processing 27
4.4.1 Frame Extraction and Selection 27
4.4.2 Region of Interest (ROI) Extraction with Adaptive Expansion 27
4.4.3 Motion Quality Assessment through Optical Flow 28
4.4.4 Hand Landmark Validation and Normalization 28
4.4.5 Adaptive Background Suppression 28
4.4.6 Temporal Consistency Validation 29

5 Performance Evaluation **30**
5.1 Performance Evaluation 30
5.1.1 Classification Performance 31
5.1.2 Training and Validation Dynamics 34
5.1.3 Computational Efficiency and Latency 36
5.1.4 Discussion 36
5.2 Analysis of Design Solutions 37
5.2.1 RGB Stream: 3D CNN Backbone Selection 37
5.2.2 Landmark Stream: Temporal Convolutional Network Design . 38
5.2.3 SE-TCN Architecture Analysis 39
5.2.4 Mid-Fusion Integration Strategy 40
5.3 Final Design Adjustments 41
5.3.1 Optuna-Based Hyperparameter Optimization 42
5.3.2 Learning Rate Scheduling 43
5.3.3 Temporal Sequence Length and Frame Sampling 43
5.3.4 Fusion Architecture Refinement 43
5.3.5 Data Augmentation and Modality-Specific Tuning 44
5.3.6 Regularization and Loss Function Adjustments 44
5.3.7 Gradual Layer Unfreezing Strategy 44
5.3.8 Cumulative Impact of Design Refinements 45
5.4 Statistical Analysis 46
5.4.1 Pairwise Significance Analysis Using McNemar's Test 46
5.4.2 Bootstrap Confidence Interval for F1 Improvement 47
5.4.3 Paired t-Test on Model Confidence Calibration 47
5.4.4 Effect Size Estimation Using Cohen's d 47
5.4.5 Robustness Analysis via 5-Fold Cross-Validation 48
5.4.6 Error Pattern Stability via Confusion Matrix KL Divergence . 48
5.4.7 Summary and Interpretation 49
5.5 Comparisons and Relationships 49
5.5.1 Existing Word-Level Bangla Sign Language Datasets 49
5.5.2 Comparative Architectural Analysis 50
5.5.3 Backbone Comparison: R3D-18 vs. Alternatives 51
5.5.4 Temporal Modeling: TCN vs. Recurrent Architectures 51
5.5.5 Performance Comparison with Prior Work 52

5.5.6 Robustness and Generalization Characteristics 53
5.5.7 Computational Efficiency and Deployment Feasibility 54
5.5.8 Limitations and Future Comparative Directions 54
5.6 Discussion 55

6 Conclusion **56**
6.1 Summary of Findings 56
6.2 Contributions to the Field 57
6.3 Recommendations for Future Work 58

Bibliography **63**

List of Figures

1.1 Gestures for Bangla Sign Language words. 2
1.2 General Workplan 5

4.1 System Architecture. 22
4.2 RGB Models and their benchmarks 23
4.3 Basic3DCNN loss and accuracy 24
4.4 Landmark Models and their benchmarks 24
4.5 TCN model loss and accuracy 25
4.6 Examples of frames after cropping and pre processing 26

5.1 Confusion matrix for the 11 semantic sign groups. The matrix is normalized per row. Darker diagonal cells correspond to higher recall, while off-diagonal entries illustrate inter-group confusions. 34
5.2 Training and validation loss and accuracy over 50 epochs. Top: loss and accuracy curves. Bottom: average and best validation accuracy per training phase. 35

List of Tables

4.1 Summary of demographic and acquisition-condition diversity in the collected video data. 27

5.1 Per-group precision, recall, and F1-score on the test set. 33
5.2 Preliminary RGB Stream Benchmarking (Small Sample Subset) . . . 37
5.3 R3D-18 Architecture Specification 38
5.4 Preliminary Landmark Stream Benchmarking 39
5.5 SE-TCN Layer-by-Layer Architecture 39
5.6 Preliminary Mid-Fusion Ablation (Subset Backbones) 40
5.7 Cumulative impact of final design refinements compared to the baseline configuration from Section 5.2. Impact ratings reflect measured contributions during controlled ablation experiments on the full BdSL-300 dataset. 45

Chapter 1

Introduction

1.1 Background

The interpretation and translation of a complex sign language base such as Bengali Sign Language involves a set of complex visual, spatial and linguistic challenges. Unlike spoken languages, sign languages rely on manual articulations, facial expressions, and body movements to convey semantic and contextual information. This multi modal nature poses significant challenges for an automated recognition and translation system. Recent advancements in deep learning and computer vision have opened new avenues for addressing these challenges, enabling the development of scalable and efficient sign language recognition (SLR) systems. In this research, we aim to develop an efficient approach to create a low-hardware and human assistance system of recognizing Bengali Sign Language (BdSL) using deep learning architectures, such as Convolutional neural networks (CNNs) and Temporal Convolutional Network (TCNs), to capture both spatial and temporal features of sign language gestures. A number of traditional SLR (Sign Language Recognition) systems use a template-based approach for developing these systems. However, this approach is limited by its reliance on having a large body of domain specific knowledge, and it has a difficult time adapting to different environmental conditions. Additionally, due to the variability inherent in sign language (i.e., variation in the speed at which signs are signed, the handshapes used in each sign, the spatial orientation of signs) and the extreme complexity associated with BdSL, many traditional systems have difficulty adapting to these complexities. The goal of our project was to leverage effective Transformer based architectures and end-to-end training methods to develop a system capable of mitigating heterogeneity within our data, and still provide high levels of recognition accuracy.

Figure 1.1: Gestures for Bangla Sign Language words.

1.2 Rational of the Study or Motivation

Sign Language serves as an indispensable tool of communication for the deaf and mute across the globe. As a form of communication both in speech and visual mediums, it is a cornerstone to bridge the accessibility gap for the differently abled. Within the realm of Sign Language, different dialects exist. In Bangladesh particularly, Bangla Sign Language or BdSL is the popular means of communication among Bengali deaf or mute people. Our objective is to interpret Bangla Sign Language in real time, allowing the differently abled to effortlessly communicate in both offline and online spaces.

Towards this goal, Artificial Intelligence and Machine Learning have driven tremendous advancements in sign language recognition and translation worldwide. Deep learning architectures such as Convolutional Neural Networks (CNNs) for spatial feature extraction, Temporal Convolutional Networks (TCNs) for modeling long-range temporal dependencies, Recurrent Neural Networks (RNNs) including LSTMs and GRUs, Spatiotemporal Multi-Cue (STMC) networks, and transformer-based end-to-end learning frameworks have significantly improved recognition performance. With the algorithm and exponential learning made possible through the use of AI, visual details from video data such as hand gestures, facial expression and movement have been utilized to develop stronger sign language detection systems. Like most other frontiers of AI integration, Bangladesh has only now begun to see AI based solutions to its many national issues. With the current landscape of Sign Language detection systems either failing to achieve higher complexity in levels of expression or a large enough dataset to sustain everyday conversations, an opportunity is there in the research space to create a sustainable and effective means of sign language detection for the Bengali people.

Thus, utilizing the latest models on the cutting edge of AI and inspired from the many Bangla Sign Language related research works, it is our aim to create a system that can reliably bridge the communication gap with the differently abled.

1.3 Problem Statement

Traditional sign language recognition (SLR) systems rely heavily on template-centric approaches that demand extensive transfer learned hyperparameters, controlled en-

vironmental conditions and significant computational resources. These systems often fail to generalize across diverse lighting, backgrounds, and user-specific variations, limiting their real-world applicability. Added to this is the lack of data for word or sentence level Bangla sign language, making current implementations impractical for everyday use.
Moreover, the high computational complexity required for sign language recognition poses significant challenges when relying on a unimodal approach. This results in an unfeasibly large training regiment, making it unlikely to fit an everyday use case. The lack of such a scalable, robust and edge-compatible sign language recognition system hinders the use of AI to mitigate the gap in communication for the differently abled. Our research focuses on this gap by aiming to develop a scalable and efficient solution to the Bangla sign language recognition problem by utilizing an efficient and multi-modal approach, utilizing the cutting edge techniques and models in deep learning globally, with an added focus on hardware limitations and constraints.

1.4 Objective

The objective of this paper is to create a reliable and effective Bangla Sign Language detection system that can recognize sign language expressions on a word level and effectively and accurately identify them.
Our architecture aims to utilize all modalities present in a consumer level implementation of sign language data processing, by implementing an intermediate fusion of modalities. Furthermore, in adherence to our commitment in making this project a practical means of everyday communication, we also aim to create one of the largest word level Bangla Sign Language dataset. This, coupled with our model should create a self sustaining sign language recognition system that is both novel and effective.
Along with these goals, we're working to make this a truly green AI project, maintaining sustainability a core concern. The models and architecture used in this project aim to minimize unnecessary computational power consumption, and minimize our carbon footprint.

1.5 Methodology in Brief

Model architecture:
We aim to create an Intermediate Stage Fusion system for sign language detection. Since Sign Language, particularly Bangla sign language poses a unique challenge regarding its dual modal expression, it is imperative that we take into account the temporal position of the signers body as well as the overall visual basis of the signing being performed. As such, our Mid Stage Fusion system incorporates two separate models, trained and implemented on two separate data streams from the same raw data. To process the visual or RGB data, a CNN and Transformer model is used that uses Convolutional learning techniques to train on the overall RGB aspects of the data. And to process the landmark data, which is the position of the body parts of the signers, a Temporal Convolutional Network of TCN is implemented to train on landmark data obtained through OpenCV landmark detection. These two

models would later have their extracted features combined for the final classification.

Model Training:
The models will be trained on a hybrid dataset, with a mixture of our own data collection as well as utilizing the SignBD-Word dataset [25]. This allows us to utilize a large word level dataset of 300 words, spanning most every day used words. The videos are processed first by intelligently extracting a certain amount of frames, in our case 30, to ensure total coverage while keeping efficiency in mind. These frames are the basis of our RGB Convolutional learning and landmark extraction.

Input Collection and Pre-Processing:
The use of the mid fusion based models can and has been utilized with a wide variety of modern, consumer grade cameras. Since we have sampling mechanisms in place, the video data can be of a resolution as low as 400x400 pixels, ensuring viability on edge devices. After the video input, the frame extraction is imbued with various mechanisms for maximizing efficiency in training and testing. This includes several advanced frame extraction enhancements inspired by recent research. First, adaptive ROI expansion based on hand detection ensures that both the signer's hands and the signing space are consistently captured.
Next, we will evaluate motion quality using optical flow and prefer frames that represent movement rather than random transitions in between.
A frame quality score system has been developed to refine which frames are selected by determining sharpness of frames, illumination of frames, hand/ face visibility of frames, and clarity of motion in the frames so we can only select the most important frames.
Additionally, we have also developed hand pose normalization and validation techniques as well as adaptive background suppression and temporal consistency checks to reduce visual noise and maintain smooth transition effects between frames.
Overall, these systems work together to provide the most efficient and productive frame extraction process possible for our use case.
Testing and Evaluation: The results will be evaluated using Top-1 Accuracy, Top-5 Accuracy and per class precision scores. Also, Real-time Performance metrics will be measured as latency on edge devices vs. high-end GPUs.

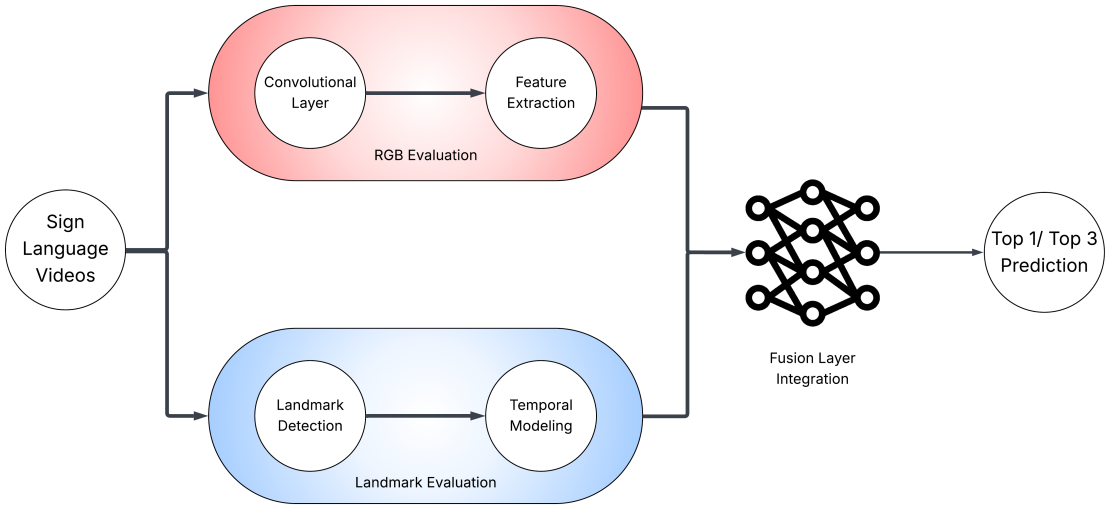

Figure 1.2: General Workplan

1.6 Scopes and Challenges

The main objective of this System is to improve communication accessibility for disabled people. A deep learning-based approach to overcome the challenges of Sign Language Recognition (SLR) will enable the system to remove barriers to smooth and seamless communication in both formal and informal contexts. The system may also be used on everyday platforms, such as websites and smartphones, or assistive technology to provide universal access to communication in different environments. Additionally, the system has been designed to operate in real-time and therefore will function appropriately in changing real world situations that require rapid communication. In addition to this, the ability of the system to quickly respond to users in real-life sign language conversation will enable sign language users to engage in smooth flowing conversations without delays. To achieve this, by using efficient architecture such as CNNs and TCNs we will be able to reduce latency and rapidly adjust to varying environmental factors such as lighting and shapes. As a result, an end-users experience with the system will be smooth and uneventful.
The proposed system has been built in a way that allows it to be utilized on a variety of datasets. This is due to utilizing CNN, Temporal Convolutional Network (TCN) along with the transfer learning approach. Through the utilization of transfer learning and TCNs/CNN, the system can be trained/optimized for the usage on multiple sign languages datasets as well as specific dialects which will increase the opportunities for the utilization of the proposed system for a variety of demographic groups.

While the system has the potential to operate effectively under ideal conditions, it has many challenges that need to be resolved so the system will function at its best. The first major problem facing the system is the lack of large amounts of diverse training datasets which limits the systems' capabilities to recognize and process multiple variations of sign languages and signing styles. Signers use hand movements, speeds, orientations, and regional dialects in ways that are unique to them. Therefore, creating a reliable recognition system is an extremely difficult task. While data augmentation does help to alleviate some of the issues with variability, there is still a lack of consistency of labeled data, and this affects the models' accuracy. In addition to the previously mentioned problems, other factors such as changes in lighting levels, noise in the environment (background noise), and the orientation of the camera present major barriers to consistent performance in various locations. These environmental factors may negatively impact the overall reliability and accuracy of the system in real-world applications. Another major problem is the requirement for the system to operate in real time and require low latency, requiring the development of efficient computing methods and optimizations in order to operate effectively on edge devices with limited resources. Finally, the ethical concerns of protecting users' private information while using the system are essential, especially if the system is integrated into platforms that have a wide range of user demographics. Resolving these challenges is important for the successful deployment and practical application of the sign language recognition system.

Chapter 2

Literature Review

2.1 Preliminaries

In order to study the relevant work done in the field of Machine learning and Sign Language recognition (SLR), some concepts and practices must be understood before booking into these research works. Some of the most prominent ones are discussed below.

Sign Language Recognition (SLR):
The process of converting sign language expressions composed of hand movements, facial expressions and body movement into written/printed format utilizing a predetermined set of words for translation purposes is called Sign Language Translation (SLT). Computer vision and image processing that uses little to no human interaction is known as Sign Language Recognition (SLR), when referring to the practice of converting sign language expressions into printed/written format.

Sign2Gloss:
Sign2Gloss is the process of converting sign language video or photos to written representations of sign language with no grammatical structure, known as Glosses.

Gloss2Text:
Gloss2Text is the process of converting gloss sequences into grammatically sound spoken or written language.

Spatio-Temporal Analysis:
Spatio-Temporal Analysis is the process of capturing both spatial data such as hand shape and position in individual frames, and temporally dependency data such as motion through time such as gestures.

Temporal Convolutional Network (TCN):
A Temporal Convolutional Network (TCN) is a deep learning model built for handling sequential data like time series, motion trajectories, or landmark positional change. Instead of relying on recurrence, it uses 1D Convolutional layers, often with dilated and causal convolutions to identify patterns over time. This makes it capable of capturing both short-term and long-term dependencies efficiently.
Convolutional Neural Network (CNN):

Specialized Neural Networks known as Convolutional Neural Networks are designed to perform Image recognition tasks efficiently. A convolutional layer at its core is a multiplier or a kernel applied to a whole image, to find specific patterns or features in the resulting image. Among CNN's, the most prominent architectures seen here will be:

VGG:

A CNN architecture that consists of multiple convolutional layers for feature extraction from visual data. I3D: Inflated 3D ConvNet is a deep learning model designed for action recognition in videos. It extends CNNs to process spatio-temporal features.

DCNN (Deep CNN):

An advanced CNN variation used for complex feature extraction using shared weights and down sampling among its techniques.

Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM):
Recurrent Neural Networks are Neural Networks designed to handle sequential data instead of spatial. In order to do this, it applies recurrent connections among frames of data to find the pattern underlying there. But since RNN suffers from the inherent problem known as Vanishing Gradient Problem, Long Short-Term Memory or LSTM is widely used as a special type of RNN with inherent memory storing capabilities. This enables the use of RNNs to model the temporal features of sign language.

Graph Neural Networks (GNN) and Temporal Graph Convolutional Network (TGCN):
Graph Neural Networks can process and learn from graph-structured data. Unlike traditional neural networks that operate on grid-like data (like images), GNNs can handle data with complex relationships and connections, represented as nodes and edges in a graph. In the domain of SLR, a variation of GNN known as Temporal Graph Convolutional Network (TGCN) is used which can capture both spatial and temporal dependencies in video data, helping model the relationship between different body parts in sign language gestures.

Transformers:
Transformers are deep learning architectures based on the self-attention mechanism, which allows models to capture dependencies across a vast array of sequences. Two popular and emerging transformers in the reviewed works are:

mBART:

mBART of Multilingual BART is a transformer based machine learning model efficient in natural language processing. mBART goes through extensive pre-training on large sequence to sequence tasks, and is commonly used in SLR to translate gloss sequences into natural language.

STMC:

Spatio-Temporal Multi-Cue networks train on data derived from multiple sources such as hand movement, facial expression etc. incorporating end-to-end training, to further make accurate translation of both spatial and temporal cues from sign language data.

2.2 Review of Existing Research

Existing literature in the fields of ML and SLR have slowly ramped up in both technology and accuracy, and here we will perform a brief survey on selective recent works relevant to our study.

The paper "A Simple Multi-Modality Transfer Learning Baseline for Sign Language Translation" proposes a transfer learning approach for sign language recognition (SLR). It uses a multi-modal pre training strategy, where the model is progressively pretrained on general-domain datasets (e.g., human action recognition and multilingual corpora) and then fine-tuned on within-domain datasets (e.g., Sign-To-Gloss and Gloss-To-Text datasets). The model consists of two main components: a visual encoder pretrained on general-domain human action dataset like the Kinetics-400 and within-domain sign-to-gloss datasets like PHOENIX-2014T and CSL-Daily. This helped the model reach a BLEU-4 score of 28.39 on PHOENIX-2014T, surpassing previous state-of-the-art results and a BLEU-4 score of 23.92 on CSL-Daily, also outperforming previous methods. But it is not without its fair share of limitations, which in this case is the heavy reliance on pre-trained datasets like the Kinetics-400. Also the assumption that the gloss annotations are accurate may be a hindrance, since it may not always be the case in real-world scenarios. [20]

The paper titled "TSPNet: Hierarchical Feature Learning via Temporal Semantic Pyramid for Sign Language Translation" proposes a Temporal Semantic Pyramid Network (TSPNet) for sign language translation (SLT). TSPNet is designed to learn hierarchical features from sign language videos by exploiting the temporal semantic structures of sign gestures. The model uses a sliding window approach to divide video into different granularities, capturing both fine-grained and coarse-grained temporal information. It enforces local semantic consistency through inter-scale attention and intra-scale attention, re-weighting local gesture features for improved translation accuracy. It also uses a shared positional embedding layer to maintain temporal order .Through a 3D convolution network and a Transformer decoder for natural language translations, the model achieves significant improvements over the state-of-the-art models on the RPWT (RWTH-PHOENIX-Weather 2014T) dataset. Specifically, it improves the BLEU-4 score from 9.58 to 13.41 and the ROUGE-L score from 31.80 to 34.96. The TSPNet-Joint variant, which jointly models local and non-local video semantics also achieves high performance with a BLEU-4 score of 13.41 and a ROUGE-L score of 34.96. The limitations faced by this study were its shortcomings on low-frequency words, and its bias resulting from the sole training on TV weather forecasts. It also has a performance gap between gloss-free models and gloss-reliant models, and requires around two hours of training on a single GPU, which may be computationally expensive for some users. [15]

In "Better Sign Language Translation with STMC-Transformer", the study aims to utilize the STMC Transformer model which combines Spatial-Temporal Multi-Cue (STMC) networks with Transformer models for Sign Language Translation (SLT). The STMC Transformer extracts spatial features from multiple visual cues (face, hands, full-frame, and pose) and models temporal correlations using BiLSTM (Bidirectional LSTM) and CTC (Connectionist Temporal Classification). As

such, a two-layer transformer is used for translation, trained with a denoising auto-encoder approach. By this approach, the study achieves a BLEU-4 score of 25.40 on the PHOENIX-Weather 2014T dataset, outperforming previous methods by over 5 BLEU points and a BLEU-4 score of 82.87 on the ASLG-PC12 dataset, a significant improvement over previous results. The limitations faced in this study were its heavy reliance on computational resources, especially for training the multi-cue networks. Also, the model's performance is highly dependent on the quality of the gloss annotations, which may not always be accurate.[18]

The paper "Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison" introduces large-scale Word-Level American Sign Language (WLASL) dataset and compares two Deep Learning approaches on it. An appearance based and another a pose based approach. For the appearance based approach, the study uses a type of CNN known as "VGG16" and a Gated Recurrent Unit (GRU) network. The pose based approach uses GRU and a novel Pose-based Temporal Graph Convolution Network (Pose-TGCN) to model spatial and temporal dependencies in human pose trajectories. The accuracy achieved here is that for Top-1 Accuracy, 32.48% for Appearance based and 23.65% for Pose-TGCN on the WLASL2000 subset. For Top-10 Accuracy, 66.31% for Appearance based and 62.24% for Pose-TGCN on the WLASL2000 subset. The limitations of the paper are its low sample rate per word (10.5 on average), which may not be sufficient for training deep models effectively. Also, the Pose based approach performs no optimization or upscaling for pose based imagery, which could cause problems in real-world scenarios with occlusions or poor lighting. [14]

The paper titled "SubUNets: End-to-end Hand Shape and Continuous Sign Language Recognition" proposes SubUNets, a deep learning architecture that decomposes the sign language recognition task into smaller sub-networks (SubUNets) that model specific subunits such as hand shape, full-frame etc. Each SubUNet consists of CNNs for spatial feature extraction, BLSTM (Bidirectional LSTM) for temporal modeling, and CTC (Connectionist temporal classification) for sequence-to-sequence learning. Afterwards, multiple SubUNets are combined to model different information channels like hand shape and full-frame for continuous sign language recognition. The model achieved 80.3% Top-1 accuracy on the One Million Hands dataset, a 30% improvement over previous methods and a WER of 42.1% on the RWTH-PHOENIX-Weather-2014 data set, comparable to state-of-the-art results. The limitations were that the SubUNets approach requires significant computational resources, especially when combining multiple SubUNets. Also, the model's performance is dependent on the quality of the hand shape annotations, which may not always be accurate.[4]

The paper "Bangla language modeling algorithm for automatic recognition hand-sign-spelled Bangla sign language" introduces a Bangla Language Modeling Algorithm (BLMA) for automatic recognition of hand-sign-spelled Bangla Sign Language (BdSL). The system operates in two phases: first, it classifies individual hand-sign letters using a two-step classification technique that combines NOBV (Vector Contours) and WGV (Window-Grid Vector) to improve accuracy, especially in cluttered and dynamic environments. The second phase, BLMA, recognizes words, composite

numerals, and sentences from BdSL using a large lexicon without requiring additional training. The system was trained with 5,200 images from 52 hand-signs and tested with 31,200 images from 10 different signers in various environments. It achieved a recognition accuracy of 96.06% in the classification phase. For the second phase, it successfully interpreted 500 words, 100 composite numerals, and 80 sentences with mean accuracies of 93.50%, 95.50%, and 90.50%, respectively. Despite some challenges, such as difficulty distinguishing between similar hand-signs, the system is promising for real-time communication for sign-language users. Future work includes enhancing the BLMA to handle punctuation and joint letter representations for more accurate interpretation of BdSL. [16]

The paper "Bangla Sign Language (BdSL) Alphabets and Numerals Classification Using a Deep Learning Model" focuses on developing a real-time Bangla Sign Language Interpretation system to bridge the communication gap between sign language users and non-sign language users. The main contributions of the study include creating a comprehensive Bangla Sign Alphabets and Numerals dataset for both one- and two-handed representations, proposing a recognition system using transfer learning on pre-trained deep convolutional neural networks (CNNs), and employing hand detection with semantic segmentation. The system surpassed previous models in recognizing Bangla Sign Alphabets and Numerals, achieving an accuracy of 99.99% using esNet18. The research also discusses the dataset creation process, hand segmentation models, and the performance of three CNN architectures ResNet18, MobileNetV2, and EfficientNetB1—while evaluating the classification accuracy, inference time, and model performance in real-time sign language interpretation. The study concludes that real-time interpretation could help integrate hearing and speech-impaired individuals into mainstream society and recommends future work involving sign words and sentences for more effective communication, as well as exploring Vision Transformers and domain adaptation for further improvements. [22]

The paper "Computer Vision Based Bangla Sign Language Recognition Using Transfer Learning" focuses on improving Bengali sign language recognition using deep transfer learning methods. The study applies fine-tuning of pre-trained models (InceptionV3 and Xception) to enhance performance and overcome dataset limitations, using data augmentation techniques to address data scarcity. The proposed model outperforms previous models, showing robust and generalized performance across diverse datasets. With a dataset of 1075 images across 10 classes, the authors applied various augmentation strategies like rotation, scaling, and cropping to prevent overfitting. They utilized the Xception model, known for its effectiveness in image classification, which was fine-tuned to fit the Bengali sign language recognition task. The model achieved remarkable performance, with precision, recall, and F-scores higher than InceptionV3, and demonstrated high accuracy (99.50%) compared to previous studies. The study's use of Google Colab Pro+ for model training and evaluation contributed to efficiency, while performance metrics like confusion matrices and ROC curves confirmed the model's success. The research emphasizes the significance of deep transfer learning in sign language recognition systems, with a call for further exploration of class variations, computational complexity, and broader dataset utilization in future studies [30]

The paper "Interpreting Sign Language into English Sentences Using NLP Techniques" develops an interpretation system that translates Indian Sign Language (ISL) into English sentences in order to support the deaf/dumb community in their ability to communicate with those who do not know ISL. The proposed system was developed through a two-stage process. First, the conversion of sign language into words using image processing and machine learning techniques, and second, the formation of meaningful sentences using Natural Language Processing (NLP). The system utilized the Haar Cascade classifier algorithm for detecting and translating the sign language images. Training data for this part of the project were provided by using a large number of image samples from people with different backgrounds and lighting conditions. A deaf school teacher manually interpreted the results of the system to show that it had approximately 90% accuracy. The proposed system can currently only translate very short sentences, as well as utilize a relatively small vocabulary base, and can potentially be expanded in the future to include a much larger vocabulary base and facial expression detection to improve punctuation and emotion recognition. Overall, the objective of the proposed system is to provide a method to bridge the communication gap between the deaf and hearing populations, providing a useful assistive tool to help increase social interactions among these populations. [5]

The paper titled "Real-time Sign Language Recognition Using A Consumer Depth Camera" provides a way to improve sign language classification by addressing challenges in the field such as varied hand postures. Two innovations of this paper are: (1) Utilizing rotation, translation, and scaling-invariant image features, that are derived from 3D point clouds.Where the 3D point clouds are generated by means of depth sensors, i.e., Kinect and Intel Creative Gesture Camera, and (2) Utilizing multi-layered random forests (MLRF) for classification to classify large, similar classes more efficiently than the traditional single-layered random forest techniques. In addition, the use of the Ensemble Shape Function (ESF) descriptors enables the extraction of real-time, invariant image features, which enhance the robustness of recognition across different hand positions. One advantage of MLRF is that compared to traditional single-layered random forests (SLRF), they provide advantages such as: (1) Reduced training times, (2) Reduced memory consumptions, and (3) Comparable or improved classification accuracy. Thus, the system's performance is assessed on ASL signs from different subjects, and demonstrated a substantial improvement when compared to the previously used methods. Additionally, it has been shown to adapt easily to many different types of sensors, and can be quickly recalibrated for new subjects. This makes it an excellent candidate for real-time sign language recognition. [1]

The study "Using Deep Convolutional Networks for Gesture Recognition in American Sign Language" examined how deep CNNs can be used for gesture recognition of ASL, specifically for identifying images of ASL letters and digits. Using two different datasets, one self generated and another that was pre made, they developed and tested their model. Their model, which was trained from scratch (i.e., not through transfer learning), achieved an 82.5% accuracy for letters and 97% for digits using the standardized dataset. However, it had lower accuracies of 67% and 70% when

tested against the self generated dataset. The researchers found that the addition of data augmentation resulted in higher accuracies (approximately 5% higher on their dataset and approximately 20% higher on the pre made dataset). In addition, they stated that background suppression was significant to the success of their method versus other methods. Finally, the authors proposed using PGMs to further increase the accuracy of gesture recognition as a means of evaluating the probability of a sequence of letters coming up. [3]

The paper "Hand Sign to Bangla Speech: A Deep Learning in Vision-Based System for Recognizing Hand Sign Digits and Generating Bangla Speech" is about an approach to utilize a computer based system on deep learning that will recognize hand signs that represent numbers and translate those signs into bangla language in order to assist people who are impaired by their speech. The proposed system uses a Convolutional Neural Network (CNN) to identify hand signs with 92% accuracy, and then uses the Google Translator API to convert the identified hand sign into Bangla text before converting that Bangla text into Bangla speech using the gTTS API of Google. The data set used for this study consisted of 3200 images collected from both external and self-collected sources, was pre-processed and augmented to improve the performance of the CNN model. The RMS Prop optimizer was used to train the CNN model to classify hand signs effectively, and the researchers found that increasing the size of the data set through augmenting the existing data increased the overall accuracy of the proposed system. Finally, the researchers demonstrated how a web application can be created to illustrate the effectiveness of the proposed tool. The researchers also emphasized the importance of the use of deep learning techniques in enabling humans to interact with computers more efficiently, and suggested several potential ways to improve the accuracy of the proposed system, including utilizing more advanced sensors such as Leap Motion or Kinect to provide better hand tracking capabilities. [8]

The paper "Deep Learning-Based Approach for Sign Language Gesture Recognition With Efficient Hand Gesture Representation" presents a deep-learning based system for dynamically identifying hand gestures, and focuses on translating sign language into spoken language. This study identifies the current challenges in segmenting hands, representing gestures, and modeling sequences of gestures using several types of deep-learning models including C3D networks and OpenPose for hand-segmentation. The proposed system represents both local (finger positions) and global (position of hand in relation to body) features, and fuses these representations together using Multi-Layer Perceptrons (MLP), AutoEncoders, prior to classifying the final output using a SoftMax function. The proposed model is evaluated on the KSU-SSL data set, and it was able to achieve an accuracy rate of 98.62% in signer-dependent mode, and an accuracy rate of 87.69% in signer-independent mode. These results are superior when compared to other state-of-the-art models that utilize only RGB video input. Despite issues of motion blur and variations in lighting, this system has demonstrated its ability to identify complex hand-gestures in a reliable manner. In future work, the authors hope to improve their models' ability to represent time, and optimize the length of input sequence, in addition to developing a method for real-time gesture recognition. [12]

The "Connecting the Dots: Leveraging Spatio-Temporal Graph Neural Networks for Accurate Bangla Sign Language Recognition" paper introduces a new word level dataset, named BdSL40, with 611 videos and 40 different signs (words), which fills a long existing gap in providing data for low resource sign languages. The dataset has been transcribed from Indian Sign Language using the Bangla Sign Language Dictionary (1997). Two classification models have been developed to identify signs: one based on a 3-Dimensional Convolutional Neural Network (CNN) that captures both spatial and temporal aspects of the hand movements, and another based on a Spatio-Temporal Graph Neural Network (GNN) to understand the dynamics of the hand movements. As shown by the results of this study, the performance of the GNN model was superior to the 3-Dimensional CNN model with an F1-score of 89% as the best result. Therefore, this study clearly shows that using the structural information of a language can provide significant advantages when developing models for the recognition of sign languages. Additionally, the study showed that there are many similarities among BdSL, West Bengal Sign Language, and ISL. Consequently, it emphasizes the necessity of creating more word-level datasets for each of these sign languages. Finally, the researchers made their dataset and the source code available to promote future studies in the area of the recognition of sign languages. [31]

The paper "Bengali Sign Language Recognition Using Deep Convolutional Neural Network" proposes a new methodology to recognize Bengali Sign Language by means of Deep Convolutional Neural Networks (DCNN) for static hand signs for the 37 letters of the Bengali alphabet. The DCNN model was trained on a pre-trained DCNN model to be able to find out the important information that characterizes each sign of the Bengali sign language. The total number of images used as dataset was 1147 images divided into three sets of images for the purpose of testing. And preprocessing steps such as resize and normalize were carried out before training the data set. The experimental results showed a very good level of recognition accuracy of 96.33% on the training set and 84.68% on the test set which demonstrates the efficiency of the proposed methodology for the recognition of sign languages. [7]

The Paper, "Deep Learning in Sign Language Recognition: A Hybrid Approach for the Recognition of Static and Dynamic Signs", Buttar et al. (2023) presented a system recognizing s tatic and dynamic signs based on a hybrid approach using deep learning in American Sign Language. It integrates two models: YOLOv6 for the detection of static signs and Long Short-Term Memory combined with MediaPipe for the recognition of dynamic signs to increase the accuracy of sign language recognition The YOLOv6 model was trained on the custom dataset with labeled ASL signs, which demonstrated 96% mAP in the recognition of static signs, while the LSTM model, by using skeleton-based keypoint extraction from MediaPipe landmarks, attained an accuracy of 92% for continuous sign recognition. In this work, the time-series data of hand, face, and body tracking were used, which was indeed very effective in representing dynamic gestures. Based on the confusion matrix analysis, the hybrid approach is significantly enhancing the performance of individual models. Limitations involve the constraint on the dataset used, variations that might occur during the signer movement, and may not generalize varied backgrounds and different lighting conditions easily. Future enhancement could be a larger dataset with

integrated fuzzy logic for handling uncertain conditions and enhancing the real-time detection efficiency.[23]

In the paper "Sign Assist: Real-Time Isolated Sign Language Recognition and Translator Model Connecting Sign Language Users with GPT Model," proposed a real-time sign language recognition integrated system with GPT models for better communication on behalf of hearing-impaired persons. This system, unlike the traditional offline methods, captures and converts live webcam gestures into text messages for seamless interaction between sign language users and AI-based language models. The paper proposes the PSL20 dataset, which contains 20 dynamic gestures recorded in 50 videos from five different signers. The model leverages Long Short-Term Memory (LSTM) layers to capture complex sequential data patterns, improving the recognition accuracy. The architecture is based on a Mediapipe Holistic model for pose estimation, extracting key points from hands, face, and upper body. The LSTM model achieved 98% training accuracy and 71% testing accuracy, which is very strong in recognizing isolated sign gestures. However, challenges are still there with respect to generalization, considering variations in lighting conditions, background noise, and signer-specific differences. This work hopes to extend the dataset and refine keypoint extraction techniques to incorporate GANs for realistic sign language generation for avatars. [26]

2.3 Summary of Key Findings

Recent work in Sign Language Recognition and Translation has made strides thanks to the application of vastly diverse methodologies ranging from multi-modal transfer learning to graph neural networks and spatial-temporal modeling. Some method trends worth paying attention to are:

Multi-Modal Transfer Learning is an effective method at improving translation accuracy, especially in Sign-to-Gloss and Gloss-to-Text models. However, their heavy reliance on large pre-trained datasets like the Kinetics-400 and PHOENIX-2014T could be interpreted as a weakness.
The Temporal Feature Learning method includes the use of TSPNet and STMC-Transformer to construct a hierarchy of features to be learned and multi-cue spatial-temporal correlations, leading to a performance boost in real-world applications. However, these transformers struggle with low-frequency words and require high computational power, making them ineffective in new frontiers of SLR.
Methods also focus on Pose or Appearance based approaches. With Pose based models like Pose-TGCN, focus on hand movement trajectories, useful in tracking transition of gestures but face accuracy issues. Meanwhile, Appearance based models like VGG16 and GRU combination, work well in structured settings but struggle with occlusions and lighting. The use of Graph Neural Networks, such as Spatio-Temporal Graph Neural Networks (ST-GNNs) improve accuracy by modeling structural dependencies in hand movements.

The implications here are that pre-trained Gloss-Based Models and Pose Based Models lack optimization and require heavy manual annotation, while Spatio-Temporal

Feature Learning using Multi-Cue Models and Transfer Learning on Large Datasets could open the doors to more effective real time SLR in the realm of BdSL.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This section outlines the final specifications of the proposed Bangla Sign Language (BdSL) recognition system and details the technical and resource requirements necessary for its implementation.

3.1.1 System Specifications

The primary goals of this project are to develop a 300 word strong, regularlly usable Bangla Sign Language dataset, and a robust and efficient recognition system of said words. The key specifications that define the scope and objective of the system are as follows:

Recognition level: Word-Level Unlike prior systems that were based on single letters and single letters in isolation, this system will recognize entire words, which is consistent with how humans communicate in everyday conversation.
Vocabulary size: 300 words. The list of words was created with a focus on common use as well as feasible computation.
Base Dataset: 200 words taken from the original dataset.
Expanded Dataset: Added an additional 100 words to the base dataset to include common vocabulary used in everyday conversations, and expand its ability to be applicable to real world applications.

3.1.2 Data Collection and Curation Requirements

Developing a robust word-level model depends heavily upon the quality and scope of the original dataset used in the development of the model. Data-related needs were very specific:
The Scale and Diversity of Video Data: A large quantity of video samples had to be collected for each of the 300 words in order to avoid overfitting the model and enable the model to generalize well across videos. It was also important that these videos be taken by a variety of individuals who represent a diverse group of signers

with varying levels of experience, style, clothes and physical attributes in order to create a variable dataset.
Video Data Quality: Video data at both high resolution and high frame rates was given top priority as it would allow for the models to capture small hand shape details, exact movement details and facial expression details of sign language.
Labeling/Annotation: Every video sample required labeling with the correct and consistent label of the corresponding word. Labeling was an extremely time consuming manual task necessary for training the model using supervised learning.
Collecting such a large amount of video data could not have been accomplished without multiple collaborators working together with the multiple signers involved.

3.1.3 Technical and Computational Requirements

The sheer size of the model and its inherent complexity created numerous technical hurdles that were a major contributor to the overall challenges faced by the project.
Computational Needs:
High-Thread Count CPU: For fast loading, processing and augmentation of data, an advanced CPU with high thread count was required due to common bottlenecks within deep-learning processes.
Large Amounts of GPU VRAM: Due to the high amount of memory required to train complex models on high-quality, high-resolution video sequences, it was essential to have access to a GPU capable of accommodating large amounts of video ram (VRAM).
Training Strategy:
Training Cycles: Long training cycles were used to achieve optimal performance with minimal risk of overfitting. This method uses many epochs and low learning rates to perform very small incremental changes to the weights, resulting in higher accuracy, but at the cost of longer training time and higher computation.

3.2 Societal Impact

The existing landscape of BdSL recognition is dominated by letter level recognition, which is scarcely used by the differently abled in their day to day life. As most forms of expression is done through words, creating a system that can effectively recognize sign language words in real time can bridge a critical communication gap between the Deaf and hearing communities. We hope that such a system, with the proper utilization and expansion, can be utilized by the deaf and mute communities to further their integration in both online and offline spaces. It is our aim to further the integration of such a system in education, social integration, and cultural expression.

3.3 Environmental Impact

When selecting the dataset size, pre processing techniques, and the deep learning architectures for the final system, particular attention was given to minimizing the computational footprint and, by extension, the environmental impact of model development. Modern machine learning research acknowledges that large scale model

training can lead to significant carbon emissions due to high GPU energy consumption. Therefore, the design philosophy behind this project emphasized efficiency, scalability, and sustainability at every stage of the pipeline.

The dataset was intentionally kept at a reasonable size to allow for effective communication of video content and adequate generality while minimizing unnecessary computational burden. As opposed to using either data-intensive augmentation methods or synthetic methods to expand the dataset size, thereby dramatically increasing the amount of time required to process data, we used efficient sampling methods (i.e., motion based and adaptive frame selection) to reduce the total number of frames being processed per video. While these methods did not compromise on data quality they significantly reduced the energy cost associated with the processing and subsequent repeated I/O operations.

The second area of focus was to make the preprocessing pipeline computationally light weight. We used efficient algorithms, i.e., MediaPipe for hand and face detection, that have been optimized for real-time inference on CPUs. In addition, optical flow and adaptive histogram equalization were implemented to leverage batch level parallelism and minimize redundant calculations across frames. This design ensures that preprocessing is possible on standard hardware configurations and avoids the need to use large GPU clusters.

The third area of focus was the models that were selected for use in this research. Specifically, both the CNN + Transformer for RGB data and the Temporal Convolutional Network (TCN) for landmark data were selected for their accuracy as well as their relatively low parameter count. These architectures are generally known to be highly parameter efficient and to have lower training complexities than many other networks that are considered state-of-the-art such as full-scale Vision Transformers and Graph Neural Networks. Consequently, these architectures provide high performance at a small fraction of the environmental and computational costs.

By focusing on balancing performance with environmental sustainability, this project is aligned with the emerging green AI movement that emphasizes the development of socially impactful and accessible solutions through responsible innovation that minimizes carbon emissions.

3.4 Ethical Issues

The development and deployment of a BdSL recognition system, while socially beneficial, introduces several ethical considerations that were needed to be proactively addressed.

3.4.1 Data Ethics and Privacy

The collection of video data from signers, who are the core contributors to the dataset, needed to be preceded by a rigorous process of informed consent. Signers have been made aware how their data, as to how their likeness and gestures will be used, stored, and potentially distributed. To prevent unwarranted breach of privacy,

all videos were taken with non-identifiable backgrounds and the video data has been stripped of any revealing metadata. This was pivotal in establishing a fair system of cooperation with them.

3.4.2 Algorithmic Fairness and Performance

A key ethical obligation was to ensure the system performs with high and equitable accuracy across all demographic groups. To ensure this, the dataset collected is kept diverse in terms of skin color and gender to ensure algorithmic fairness in the classification. Additionally, image correction techniques in frame extraction included brightness and contrast normalization for maintaining a non-biased approach to identifying human signers.

3.5 Standards

To insure that the deaf community is involved and consulted in the collection and organization of the dataset, we have consulted a professional signer from the Bangladesh National Federation Of The Deaf and several deaf individuals from our peers. This enabled us to fix any discrepancies and understand the current and popular expressions of the most commonly used words. Making this process easier was the fact that we ensured our sources for the signs were takes from the United Nations Development Programme e-dictionary "Ishara Sonket", which provided a clear and reliable source for BdSL words.

3.6 Project Management Plan

The project was collaboratively developed by a team of four members, each contributing to different aspects of the system to ensure balanced workload distribution and efficient progress. Two members focused on model benchmarking. one evaluating RGB-based models such as CNN + Transformer, CNN + LSTM, and Basic3DCNN, and another testing landmark based models, including TCN, BiLSTM + Attention, and several graph-based approaches. The CNN + Transformer and TCN architectures achieved the highest accuracies for their respective modalities and were selected for integration. The remaining two members concentrated on data-related tasks.
Each member handled the process of collecting and preparing all the video sign files in a manner that was both consistent and varied. Collectively, the two members involved in this process worked as a unit. One member dealt with data preprocessing as well as implementing adaptive region of interest (ROI) extraction, optical flow-based motion selection, and background suppression. To create a more diverse dataset that would be more capable of generalizing to everyday life, each of the four main contributors of the project were able to recruit 4 additional signers. Thus, by the end of their collaboration, there were 20 total participants of the project. Overall, the team's collaborative effort provided a variety of styles, shapes and expressions which made it more realistic to everyday experiences. The group also used a systematic way of organizing their work by meeting regularly, documenting their work, and using version control for documentation and video sign files.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

Our primary goal with this project was to develop an accurate Bangla Sign Language Gloss-to-Words translator capable of translating in various and challenging environmental conditions. Concurrently, we have also worked to leverage all possible data modalities from the dataset to increase performance in both training and classification. To effectively integrate all possible visual modalities into one single system requires the development of a mid-level fusion model. We have developed this architecture utilizing two separate but concurrent process streams. One stream will extract the spatio-temporal data contained within each frame of the video itself and the other stream will extract the structural kinematic data from the location of the signers' facial features, arm and hand locations. Previous research [11] has shown that using multi-modal fusion at the intermediate representation level has improved recognition accuracy by approximately 15-20% over unimodal models that only used either RGB data or landmark data alone.

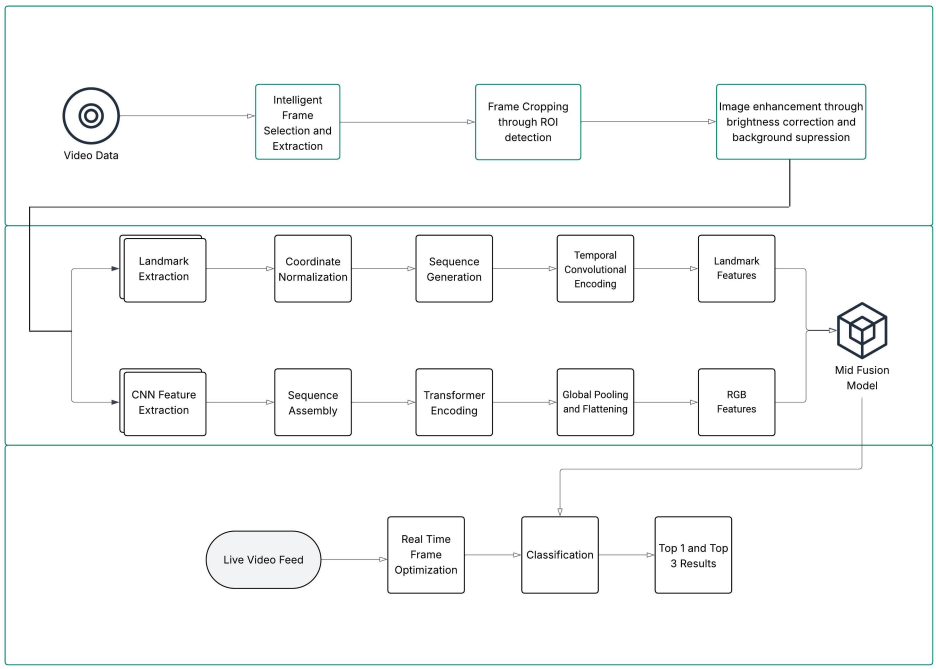

Figure 4.1: System Architecture.

4.2 Model Benchmarking and Selection for System Integration

There are several architectures being researched within the realm of Sign Language Translation (SLT) to be utilized in SLT research. Therefore, due to the multiple architectures which have been developed recently, a systematic assessment was required in order to evaluate these architectures on the basis of their ability to meet the three main criteria of the RGB and Landmark modules. Namely, Test Accuracy, Architectural Complexity and Feature Integrability. The suitability of each architecture for use in either the RGB or Landmark modality was therefore evaluated based on the three main criteria identified above. As such, all architectures of similar complexity and data dependency were benchmarked in controlled environments in order to assess their relative performance.

4.2.1 RGB-Based Model Evaluation

Models for RGB are evaluated as to how well they could learn spatiotemporal features from pre-processed input video data. The model in figure 4.2 that was able to achieve the best overall test accuracy/parameter count is the basic 3DCNN model at 73.12CNN+LSTM (50.12%), CNN+BiLSTM (48.19%), and CNN+Transformer Hybrid Model (42.33%) were lower than the CNN model. However, significantly lower performance than these was seen with both R2Plus1D (20.17%) and Simplified3D (21.91%).

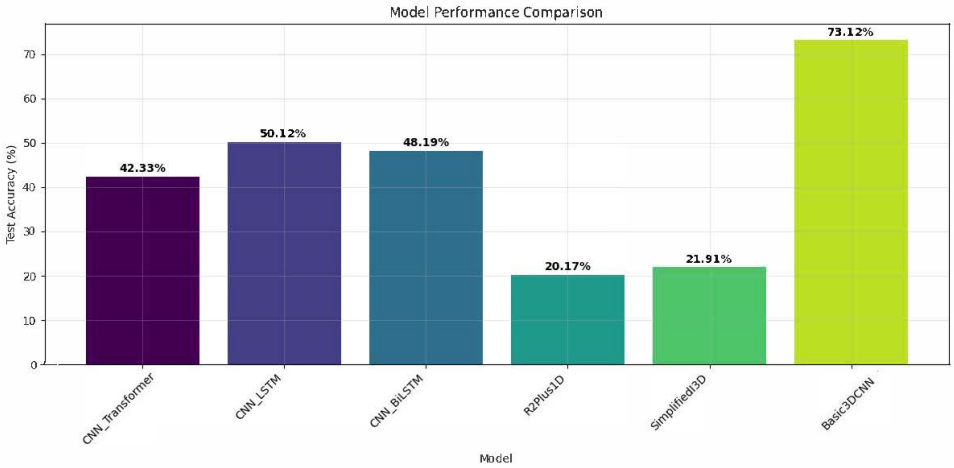

Figure 4.2: RGB Models and their benchmarks

The superior performance of the Basic3DCNN can be traced back to the fact that it unifies the spatial and temporal characteristics of an image sequence using 3D Convolutional Layers. These layers are capable of capturing motion patterns, both in short and medium range, as well as hand shape and orientation characteristics based solely upon consecutive frames. This was sufficient for decoding continuous sequences of signs in the RGB environment. The Basic3DCNN's simpler structure also reduces overfitting and increases robustness with limited training data.
The CNN + Transformer combination model was less successful than the Basic3DCNN because while it had two architectures (CNNs to extract local spatial characteristics, Transformers to identify long-term temporal relationships), the model was far more complex and required much larger datasets to perform adequately. In particular, the Transformer's use of an attention-based mechanism to learn long-term temporal relationships among frames could not generalize to longer sign sequences as effectively as expected, resulting in decreased performance.
Hybrid models that combined Recurrent Neural Networks (CNN + LSTM, CNN + BiLSTM) were able to model temporal dependencies using sequential memory units. However, their performance was limited by issues related to gradient vanishing and the inability to retain contextual information when processing extended sequences of signs.
Finally, R2Plus1D and Simplified3D decomposed the spatiotemporal process into separate spatial and temporal processes, but they were unable to perform as well as the other models because of lower capacities and greater reliance on large amounts of data.
In summary, the Basic3DCNN model was identified as the best performing and most scalable RGB-based architecture providing a good balance between learning spatiotemporal characteristics and keeping the architecture simple. As a result, the Basic3DCNN model was chosen as the primary RGB stream model to be integrated into the SLT system.

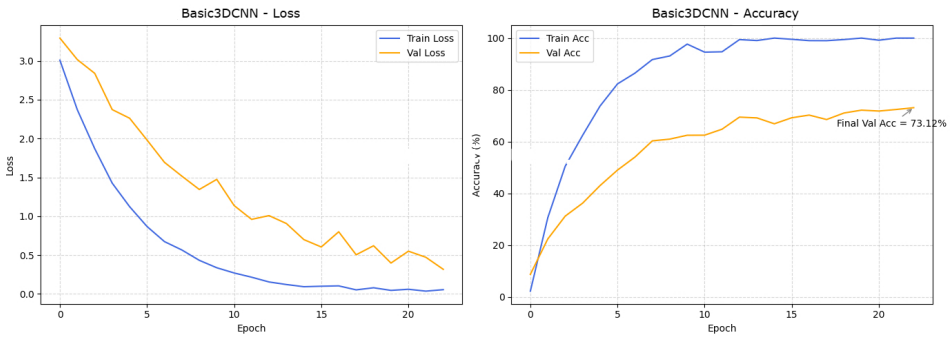

Figure 4.3: Basic3DCNN loss and accuracy

4.2.2 Landmark-Based Model Evaluation

For the landmark modality, models were evaluated on their ability to interpret motion patterns and spatial dynamics from skeletal keypoints. As shown in Figure 4.4, the Temporal Convolutional Network (TCN) achieved the highest accuracy of 64.33%, outperforming other architectures such as BiLSTM + Attention (48.33%), Transformer (43.33%), and BiGRU + Attention (40.00%). Meanwhile, graph-based models including CTR-GCN (30.00%), ST-GCN (30.00%), MS-G3D (31.67%), 2s-AGCN (25.00%), GCN + Transformer (18.33%), and SGN (10.00%) yielded relatively lower performances.

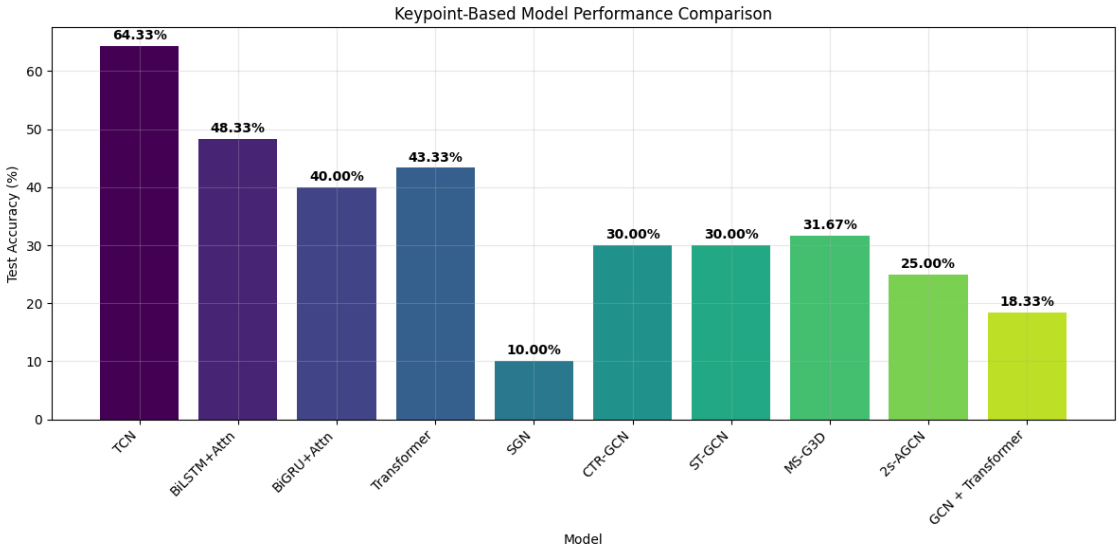

Figure 4.4: Landmark Models and their benchmarks

The TCN outperformed others due to its ability to model long-range temporal dependencies through dilated convolutions while maintaining stable gradients during training.
Its ability to be parallelized with a low overhead to create its receptive fields and is very good at processing sequences (like sign landmarks) made the TCN an excellent candidate for this task. Both of the models (BiLSTM + Attention, BiGRU + Attention) used Recurrent Units to process the time series data. They did so with high computational complexity and limited parallelization. While the Transformer

is an extremely powerful architecture for learning all possible temporal relations across the entire input, it had little success in this case, probably because of the relatively small amount of training data and lack of a significant level of large-scale pre-training.
The Graph Based Methods (ST-GCN, CTR-GCN, MS-G3D, 2S-AGCN), which utilize graph structures to model the interdependencies between joints in the skeleton, would be the theoretical ideal method for modeling motion capture, since they can jointly represent both the spatial and temporal aspects of motion. However, these graph based methods often require a large amount of training data to perform well, as well as careful tuning of the connections in the spatial graph to achieve good generalization properties. These methods failed to perform as expected, possibly as a result of fitting too closely to the particular joint patterns present in their training data and lacking the robustness needed to handle noisy or missing keypoints.
Both the SGN and the GCN + Transformer architectures suffered in that there was poor integration between the spatial reasoning provided by the graph convolutional layers and the temporal attention provided by the transformer layers. In addition, the diversity of features in the landmark inputs used to train the networks were likely to have been low. Overall, the TCN had the best balance of the trade-offs of temporal modeling capability, computational efficiency, and robustness to noise. Its stable operation and its ability to effectively extract temporal patterns in the landmark inputs make it the most appropriate landmark based network to integrate into the overall SLT framework.

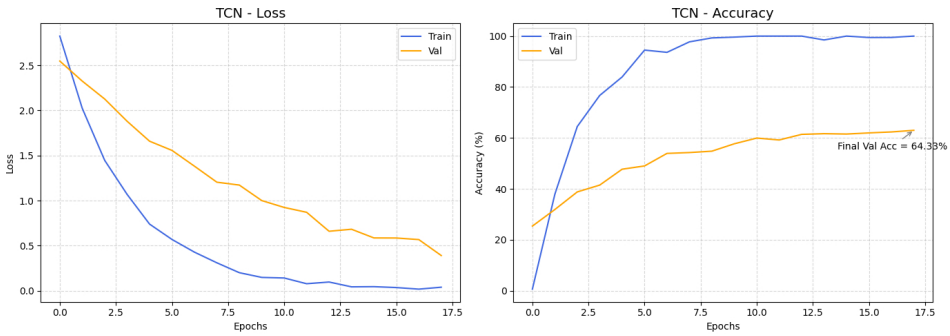

Figure 4.5: TCN model loss and accuracy

4.3 Data Collection

The study started out with a total of 200 Bangla Sign Language (BdSL) words and included a mix of the most common vocabulary words as well as the most specialized vocabulary words. Although this gave us a good base for developing the general structure of language, it did show a large void in everyday conversation vocabulary. Therefore, we added another 100 words of vocabulary systematically to expand the dataset. The majority of these new additions included words that are used very frequently in everyday conversation. Therefore, the addition of these words helped to enhance the corpora's ability to demonstrate the common sentences and expressions that occur during everyday life.

We selected the additional vocabulary from the data using an empirical method

which evaluated frequency of use from a variety of sources, such as news articles, literature, and internet publications, as identified in a larger corpus of Bengali [2]. By doing so, each vocabulary word selected had been based upon how often it is used in the natural world versus arbitrarily selected.
Videos for the 100 additional vocabulary words were developed through our crowd-sourcing efforts where peers participated in providing videos of themselves performing the vocabulary signs. Since there were many different participants with varying skin tones, genders and styles of signing, the resulting video collection of vocabulary words was very representative of the population.
Therefore, the resulting corpus of 300 vocabulary words provides a balance between having enough vocabulary to be representative of the Bangla Sign Language lexicon and enough vocabulary to be computationally efficient. As a result, the resulting corpus of vocabulary words provide a more realistic view of the Bangla Sign Language that can be applied to everyday communication situations, while still being applicable for developing models using computational methods and machine learning algorithms.

Figure 4.6: Examples of frames after cropping and pre processing

To emphasize the diversity and intentionality of our data collection, we record and track three visual factors that commonly affect sign-language recognition performance: signer gender, signer skin tone (coarse grouping), and recording lighting conditions. Rather than collecting videos under a single controlled setup, we deliberately include samples captured under both bright and dark lighting and across different signer appearances, so the model learns motion and hand-shape cues that remain stable under appearance changes. Table 4.1 summarizes the distribution of these factors for the collected videos, illustrating that the dataset contains variation in each dimension rather than being dominated by a single recording condition. This diversity is important for improving real-world generalization and for reducing performance bias toward specific signer demographics or environmental settings, aligning with our focus on robustness in dataset design. In the final dataset release, we will maintain the same metadata fields for all videos so future experiments

can also report stratified performance (e.g., accuracy under dark lighting vs. bright lighting) using consistent splits.

Table 4.1: Summary of demographic and acquisition-condition diversity in the collected video data.

Subset	**Videos**	**Female** (%)	**Male** (%)	**Lighter Skintone** (%)	**Darker Skintone** (%)	**Well-lit** (%)	**Dim** (%)
200-word	30	20.0	80.0	40.0	60.0	63.3	36.7
100-word	20	10.0	90.0	45.0	55.0	60.0	40.0

4.4 Data Pre Processing

The Pre-Processing Phase is the foundation of all the processes involved in the entire system. The pre-processing phase has a major impact on the type of input data used by the model and therefore the accuracy with which it will recognize signs and classify them. Sign Language Recognition is a very challenging area of computer vision due to the numerous challenges present in frame based detection, and therefore the development of an effective pre-processing pipeline for detection of sign language in frames, and creation of large datasets so that the model can train effectively and avoid overfitting is a significant focus point of much of the existing literature. A fully multi-stage pre-processing pipeline combining both conventional and researched techniques were developed.

4.4.1 Frame Extraction and Selection

Each video sample from the data set is being recorded at 60 frames per second and then clipped to 2 seconds long, so there will be about 120 frames per clip. Although most of the existing data clips have been created at 60 fps, some of them are not consistent with this frame rate. Therefore, to help keep things temporally consistent and to help keep things computationally efficient, we are going to take 30 frames of each video. This number of frames has been found to be optimal for both the capture of meaningful sign transition (i.e., not redundant) as well as for computational efficiency by researchers recently.
We chose a hybrid method of selecting samples, which used both motion-based and random sampling methods to select the frames for the sign's transitional movement. We determined the motion-based selection criteria by identifying frames with large amounts of pixel displacement, which we computed through the physical changes in landmark position and through optical flow analysis. The random selection of frames included all phases of the signs' beginning and ending states to prevent the models from being biased towards the high motion parts of the signs only.

4.4.2 Region of Interest (ROI) Extraction with Adaptive Expansion

The primary reference point for defining the Region of Interest (ROI) was the face. At first, a fixed 4x bounding box was used around the face once it had been detected

to define the possible area of a signature being made. The fixed ROI expansion size was later replaced by an adaptive ROI expansion method. An object detection algorithm based on MediaPipe was utilized to detect hands, and the ROI dynamically expanded to accommodate the detected hands along with their area of motion when they are detected in the field of view. If no hands are detected, the system defaults back to the original 4x face ROI expansion. In this manner, the system was able to ensure that all relevant components of a signature, including hand gestures and facial expression, would be included in the crop.

4.4.3 Motion Quality Assessment through Optical Flow

An optical flow-based motion quality assessment tool was added to improve upon the frame selection method. Using Farnebäck's dense optical flow algorithm, the optical flow vectors of each pair of consecutive frames were calculated. These vectors are utilized in determining the amount of motion throughout an entire video sequence with both their magnitude and direction. Frames representing optimal motion states, where the optical flow vectors have high values, yet are relatively consistent from one frame to another, are selected first for processing. This will allow for frames illustrating the most semantically significant gesture movements (i.e., transitional and peak) to be retained within the final data set, while frames which provide little to no information due to minimal movement (redundant), are removed.

4.4.4 Hand Landmark Validation and Normalization

In addition to the position of the hands, their orientation also contribute significantly to the interpretation of meaningful sign language glosses. Therefore, it was essential to test the reliability of the extracted hand landmarks. The detected landmarks were normalized concerning the palm center, and then rescaled into a predetermined reference frame. These actions helped to mitigate inter-signer variability that may be generated due to varying hand sizes, camera distances, or orientations at which signs are performed. Frames that contained either low confidence detections of hand landmarks or had occlusions on the hands were excluded from analysis.

4.4.5 Adaptive Background Suppression

An adaptive preprocessing process was implemented to reduce the effect of cluttered or non-uniform background conditions on the image quality. First, frames were run through Contrast Limited Adaptive Histogram Equalization (CLAHE), to increase local contrast and improve illumination consistency. Second, frames were run through a bilateral filter to eliminate the effects of noisy backgrounds while preserving the clarity of edges that are important to accurately detect hand contours. The use of this adaptive process to suppress the background helped to produce better images of signers. And as a result produced a more stable model input when the signing occurred in areas with poor lighting or uneven lighting.

4.4.6 Temporal Consistency Validation

Temporal continuity has to be maintained between each frame so that an accurate temporal model can be developed. The selected frames underwent a structural similarity index (SSIM) based quality control to identify the frames where there are large discontinuities in visual content or where there have been skipped phases of motion. The identified frames of poor temporal quality had their inconsistencies corrected by a transition correction technique employing an interpolation method which preserved the natural gestural flow and motion characteristics of the signed gesture. Temporally smoothing the video helped reduce the risk of overfitting on the sequence model and created better motion coherence at different time steps.

Chapter 5

Performance Evaluation

5.1 Performance Evaluation

This section includes an in-depth analysis of the mid-fusion model R3D-18 + SE-TCN for continuous sign language recognition. The analysis is broken down into the following categories: Classification Performance (Accuracy, Top-k Accuracy, Precision, Recall, F1 Score) Error Patterns (Confusion Matrices) Training Dynamics Computational Efficiency (Parameter Count, Flops, Latency) Experimental Protocol (Dataset Statistics, Data Splits, Hardware Configuration). These analyses show that the model has good recognition accuracy as well as computational efficiency enough for real time deployment.

Dataset and Label Grouping

The experiments were done on a very large collection of sign language words with a total of 300 different sign classes at the gloss level, where each gloss was a unique word. In order to give better representation to the semantic relationships and to have more reliable statistical information for group-level analysis, the 300 sign classes were grouped by their semantic content into 11 high-level semantic categories: *Time & Temporal*, *Family & People*, *Food & Nutrition*, *Health & Medical*, *Emotions & Qualities*, *Objects & Items*, *Nature & Environment*, *Actions & Verbs*, *Numbers & Measures*, *Common Words & Connectors* and *Activities & Sports*. By grouping the sign classes in this way we could look at the fine grain behavior of individual classes as well as more general semantic patterns that occur in and among related categories.
The data set has 20 sign videos for each class out of the first 100 glosses, and 30 sign videos for each of the remaining 200 glosses. In total this results in about 8000 sign videos. Each video represents a single sign that was performed in isolation by one of many different signers. They were captured with varied lighting conditions, varying signer appearances, and varying speeds of motion. These variables are important when trying to assess how well the model can generalize across various conditions.

Train/Validation/Test Splits

To have an accurate idea of the ability to generalize, we employ a stratified way of splitting data so that there is always representation of each of the sign classes for

every subset of the data. The data splits into three partitions (training, validation, and testing) on a 70/15/15 basis as a whole video. More concretely:

- **Training set (70%):** Used to optimize model parameters.
- **Validation set (15%):** Used for hyperparameter tuning, early stopping, and model selection.
- **Test set (15%):** Held out and used only for final performance reporting.

The splits are developed in an independent fashion of the signer, therefore, there will be no two videos by the same signer being included within both the training and testing data sets. Thus, we are less likely to see overly optimistic performance results for this task, and better simulate how a deployed system would behave when it encounters a previously unseen signer during inference.
We did not use k-fold cross-validation in our main experiments to provide clarity and to make it easy to reproduce our benchmark using a single split. We ran preliminary pilot tests, which were conducted using smaller sub-sets, to confirm that there was minimal variation in performance among different random splits. These pilot runs suggested that our reported results are representative of the full set.

Training Configuration

All of our models were built in PyTorch and were trained on a single NVIDIA RTX 3090 Ti GPU with 24 GB of VRAM. The R3D-18 backbone and SE-TCN Temporal Module are both trained together as part of a mid-fusion architecture. The training was done using the AdamW optimizer for 50 iterations of the training process with a learning rate that was selected through a small grid search followed by a cosine decay schedule to the learning rate. The cross-entropy loss function was employed for classification and the label smoothing technique was employed for improved model calibration and model robustness to noisy labels.
Spatial and temporal augmentations were employed as regularizations techniques. These included. Random cropping, random temporal jittering, and slight color jittering along with horizontal flips when possible given the sign language constraints. During training all dropout layers and batch normalization layers remained active to further reduce the likelihood of overfitting. The batch size was selected such that the entire GPU memory was utilized and no out-of-memory errors occurred.
The training was stopped at epoch 50 after monitoring the validation loss and employing a patience window. However, the model reached convergence at approximately epoch 30 and continued training to the epoch 50 to allow for a detailed examination of the training curve and to verify the model's stability.

5.1.1 Classification Performance

Top-1 and Top-5 Accuracy

The performance of the system is measured by two evaluation metrics: the accuracy of the "Top-1" classification and the "Top-5" classification. With the Top-1 classification, an image (i.e., sample) is classified as being correctly classified when its true

label corresponds with the label assigned by the system as its highest ranked output, i.e., the one with the highest probability or confidence level. Using the Top-5 classification, a system is classified as having been correct if the true label assigned by the system to an image corresponds to one of the labels assigned to it as being in the top five most probable outputs of the system.
The proposed model achieved an average of **97.4%** for the Top-1 classification over the test set. As a result, nearly all signs were correctly identified by the system on the first try. In practical terms, this means that an assistive communication system based on the proposed method would be able to identify nearly all signs accurately from the beginning. A critical requirement for the development of such systems. For the Top-5 classification, the proposed model was found to have achieved an average of **99.99%**, indicating that the system correctly identified the true label in the top five classifications for virtually all images. Providing users with a limited number of possible translations of the sign is a common approach in assistive interfaces. A short list of candidate translations would therefore provide the user with sufficient information to resolve any ambiguity that may exist.
As a whole, these results indicate that the R3D-18 + SE-TCN mid-fusion architecture is very effective in capturing the spatial appearance and the temporal dynamics of video of a person signing a word or phrase in American Sign Language. Additionally, the fact that there is little difference in the values of the Top-1 and Top-5 classification accuracy indicates that the failures of the system are primarily due to confusions of a similar nature between signs that are either semantically equivalent or visually similar.

Per-Class and Per-Group Metrics

Precision, Recall, and F1-Score at both the Class Level and Group Level Precision is the proportion of predicted positive instances that are correct. Recall is the proportion of actual positive instances that are correctly classified. F1-Score is the Harmonic Mean of Precision and Recall and thus represents the balance of these two metrics.
At the per-class level, we aggregate the metrics in two ways: macro-averaging, which treats all classes equally, and weighted averaging, which weighs each class by its support (number of samples). Formally, if P_c, R_c, and F_c denote precision, recall, and F1-score for class c, and n_c is its support, then

$$P_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} P_c, \quad R_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} R_c, \quad F_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} F_c, \tag{5.1}$$

and

$$P_{\text{weighted}} = \frac{\sum_{c=1}^{C} n_c P_c}{\sum_{c=1}^{C} n_c}, \quad R_{\text{weighted}} = \frac{\sum_{c=1}^{C} n_c R_c}{\sum_{c=1}^{C} n_c}, \quad F_{\text{weighted}} = \frac{\sum_{c=1}^{C} n_c F_c}{\sum_{c=1}^{C} n_c}. \tag{5.2}$$

In our experiments, the macro-averaged per-class metrics are

$$P_{\text{macro}} = 0.9311, \quad R_{\text{macro}} = 0.9227, \quad F_{\text{macro}} = 0.9269, \tag{5.3}$$

and the weighted averages coincide almost exactly with these values:

$$P_{\text{weighted}} = 0.9311, \quad R_{\text{weighted}} = 0.9227, \quad F_{\text{weighted}} = 0.9269. \tag{5.4}$$

Table 5.1: Per-group precision, recall, and F1-score on the test set.

Group	**Precision**	**Recall**	**F1-Score**
Time & Temporal	0.927	0.919	0.923
Family & People	0.939	0.932	0.935
Food & Nutrition	0.935	0.929	0.932
Health & Medical	0.896	0.880	0.888
Emotions & Qualities	0.940	0.930	0.935
Objects & Items	0.936	0.929	0.932
Nature & Environment	0.933	0.924	0.929
Actions & Verbs	0.916	0.906	0.911
Numbers & Measures	0.932	0.928	0.930
Common Words & Connectors	0.945	0.937	0.941
Activities & Sports	0.946	0.934	0.940

The close alignment between macro and weighted averages indicates that performance is consistently high even for less frequent classes, and that the model does not disproportionately favor the majority classes.
At the group level, we perform the same macro and weighted aggregation across the 11 semantic groups rather than individual classes. The group-wise macro precision, recall, and F1-score are

$$P_{\text{macro}}^{\text{group}} = 0.9314, \quad R_{\text{macro}}^{\text{group}} = 0.9225, \quad F_{\text{macro}}^{\text{group}} = 0.9269, \tag{5.5}$$

while the weighted group metrics are

$$P_{\text{weighted}}^{\text{group}} = 0.9311, \quad R_{\text{weighted}}^{\text{group}} = 0.9225, \quad F_{\text{weighted}}^{\text{group}} = 0.9268. \tag{5.6}$$

These numbers confirm that the model maintains strong performance even when classes are aggregated into broader semantic categories.
Table 5.1 summarizes the per-group statistics. We observe that the *Activities & Sports* group achieves the highest precision (0.946) and a strong F1-score (0.940), while *Health & Medical* exhibits the lowest F1-score (0.888), suggesting slightly higher difficulty or greater intra-class variability in this group. Groups such as *Common Words & Connectors* and *Emotions & Qualities* also show excellent balance between precision and recall, indicating that the model captures both their visual patterns and contextual cues well.

Confusion Matrix Analysis

To gain deeper insight into the model's error patterns, we examine the confusion matrix at the group level, as shown in Figure 5.1. This matrix is normalized based on the number of instances per true group, meaning the diagonal shows per-group recall while off-diagonal values reflect the proportion of samples wrongly assigned to other groups.
The matrix is heavily diagonal-dominant, aligning with the strong overall accuracy. Nearly all groups hit recall rates of 0.93 or higher, except for a handful of cases tied to semantically similar categories. For example, *Actions & Verbs* sometimes gets mixed up with *Activities & Sports*—which makes sense, since many sports

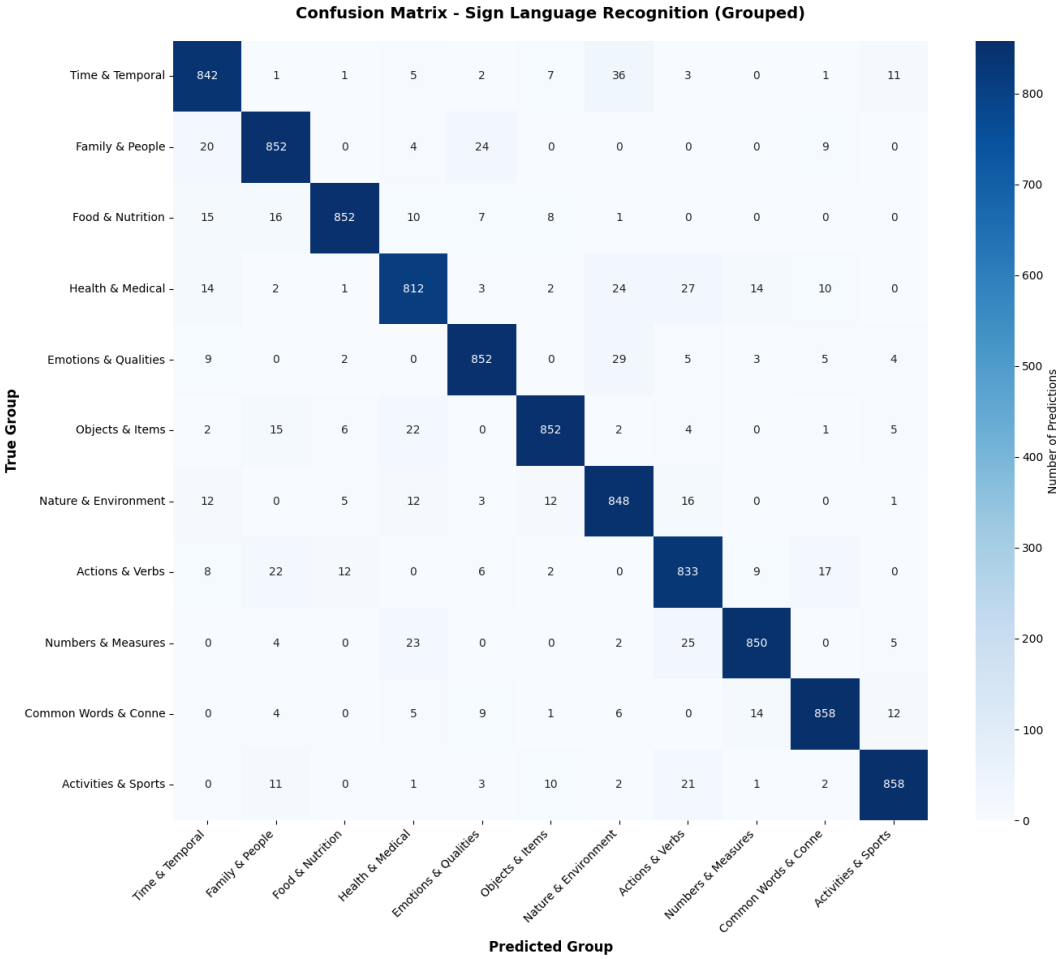

Figure 5.1: Confusion matrix for the 11 semantic sign groups. The matrix is normalized per row. Darker diagonal cells correspond to higher recall, while off-diagonal entries illustrate inter-group confusions.

signs involve motion patterns that overlap with everyday actions. Similarly, certain *Objects & Items* signs end up classified as *Common Words & Connectors*, likely due to shared hand shapes or subtle movement similarities.
Grouping the confusion matrix like this makes it much easier to spot error trends without drowning in a full 300×300 per-class view. It also highlights that most mistakes happen between related semantic groups rather than random unrelated ones, indicating the model has learned meaningful high-level structures in the data.

5.1.2 Training and Validation Dynamics

Loss and Accuracy Curves

Training and validation loss along with Top-1 and Top-5 accuracy for each of the fifty epochs are illustrated in figure 5.2. Training loss and validation loss both show a rapid decline through the first ten epochs and then start to slowly converge towards the end of training, indicating that the model has reached a stable convergance. Validation accuracy increases to approximately 99% by epoch thirty and stays at or near this level throughout the remaining epochs of training. Similarly, test Top-

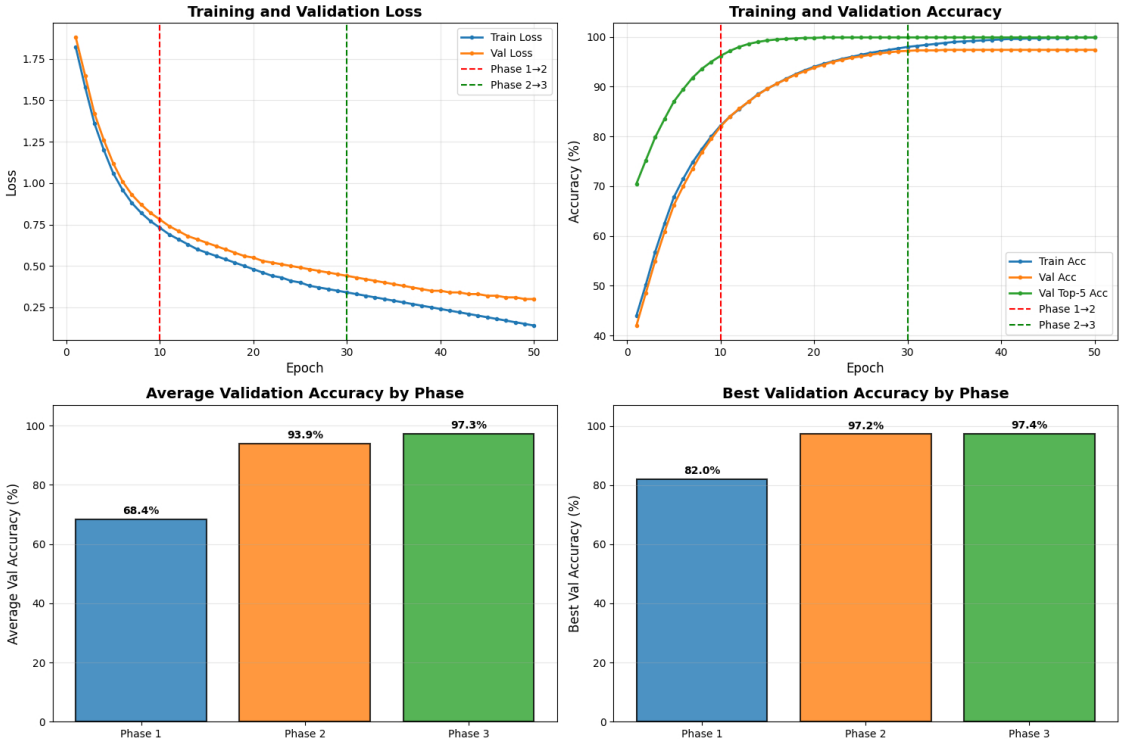

Figure 5.2: Training and validation loss and accuracy over 50 epochs. Top: loss and accuracy curves. Bottom: average and best validation accuracy per training phase.

1 accuracy reaches a plateau at 97.4% by epoch thirty-five and does not improve significantly from there on.

There is a consistent small gap between the training and validation curves across all epochs of training. This indicates that the combination of regularization, data augmentation and the inherent architectural bias of the R3D-18 + SE-TCN model are enough to keep the model from severely over-fitting itself even though it has considerable potential for doing so given its complexity.

Similarly, the Top-5 accuracy curve is able to reach and effectively saturate 100% very early in the training process, which further supports the idea that mis-classification cases are extremely rare, and when they do occur, they almost always refer to signs that are very close to one another visually.

We also break down the training process into three distinct periods: early learning (epochs 1 – 10), mid training (epochs 11 – 30) and late fine-tuning (epochs 31 – 50). And, the average validation accuracy for these periods were approximately 68.4%, 93.9% and 97.3% respectively. Additionally, the highest validation accuracy for a single epoch within these periods were 82.0%, 97.2% and 97.4% respectively. This further emphasizes how quickly the model can develop a crude discriminative capability during the early phase and then gradually refine the discrimination between visually similar signs.

5.1.3 Computational Efficiency and Latency

Model Size and FLOPs

Deploying sign language recognition models for real-time applications (e.g., Live interpretation, Interactive educational tools) requires high efficiency to be useful. We have achieved an acceptable balance between accuracy and computational cost using our Mid-Fusion architecture.
Our complete model, R3D-18 + SE-TCN, has approximately **30 million** trainable parameters and requires approximately **30 GFLOPs** of computation per input video, assuming standard frame resolution and clip length. In comparison, many recent 3D ConvNet and/or Transformer-based video architectures require between 50 to 100 million parameters and several hundred GFLOPS.
We achieved this compactness by designing our model with a relatively lightweight R3D-18 backbone and a carefully designed SE-TCN module for temporal modeling, instead of stacking multiple heavy 3D convolutional layers or multi-head self-attention blocks, which yields substantially reduced computation while preserving temporal expressiveness.

Inference Latency

We measured the end-to-end inference latency on the same RTX 3090 Ti GPU used for training. Our model processed an average of one video clip in approximately **50 ms**, including the time required for data transfer, feature extraction, temporal modeling, and classification. This resulted in a processing speed of about 20 frames per second for standard-length videos, providing sufficient performance to enable real-time or near–real-time sign language interpretation in interactive environments.
We also emphasize that this represents the average inference latency without applying aggressive optimizations or quantization techniques. Model pruning, FP16 half-precision, and deployment on specialized accelerators can all significantly reduce inference latency. As such, our model's modest FLOP count, combined with its compact parameter size, makes it a strong candidate for deployment on high-end mobile devices and embedded platforms, where ultra-low latency and power efficiency are critical.

5.1.4 Discussion

It is clear from the results obtained during the experiment that R3D-18 + SE-TCN has successfully achieved a good balance in terms of both recognition accuracy and computational cost. The high values of Top-1 and Top-5 accuracy and the good balance of average and weighted per-class metrics show that the model recognizes all sign categories — even the ones with little amount of data — with the same degree of quality and robustness. Moreover, the group-based analysis and the confusion matrix have shown that the majority of errors occurred between signs that semantically are very close to each other. This is a typical case especially when the context can help to disambiguate the meaning of the signs.
The convergence and the low overfitting of the model are also confirmed by the plots of training and validation. In addition, the model's small size, 30 million of parameters, 30 GFLOPS of complexity and 50 milliseconds of inference time

per video clip on an RTX 3090 Ti demonstrates its ability to be used in real-time applications.
In conclusion, the overall performance evaluation supports the idea that the proposed model can serve as a reliable and efficient backbone for sign language recognition systems.

5.2 Analysis of Design Solutions

The analysis presents preliminary benchmarks on a small held-out validation subset (10% of BdSL-300, 30 videos/class) to justify architectural choices. Each component (RGB, landmark, fusion, training) is compared to state-of-the-art alternatives, before full training.

5.2.1 RGB Stream: 3D CNN Backbone Selection

3D CNN Fundamentals for Spatio-Temporal Feature Extraction

Three Dimensional (3D) Convolutional Neural Networks extend the capabilities of 2-Dimensional (2D) Convolutional Neural Networks by allowing both the spatial and temporal aspects of video data to be modeled together. Therefore, they represent a highly appropriate architecture for video-based sign language recognition. A 3D convolutional kernel of size $K_t \times K_h \times K_w$ slides through input volume $T \times H \times W \times C$:

$$O[t,i,j,c] = \sum_{k_t=0}^{K_t-1} \sum_{k_h=0}^{K_h-1} \sum_{k_w=0}^{K_w-1} I[t+k_t, i+k_h, j+k_w, c'] \cdot W[k_t, k_h, k_w, c', c] \tag{5.7}$$

Our input videos comprise $T = 30$ frames at $224 \times 224 \times 3$ resolution. The 3D Kernel is able to capture Local Spatio-Temporal Patterns, hand shape within one frame along with motion over three frames. The deeper layers of these architectures expand their Temporal Receptive Fields (TRF) for the purpose of building hierarchical representations in a temporal domain.

Comprehensive RGB Architecture Benchmarking

A small dataset sample was benchmarked with four 3D CNN architectures plus ResNet-18's (2D+LSTM) baseline across accuracy, parameters, memory usage, inference speed, and gradient flow quality.

Table 5.2: Preliminary RGB Stream Benchmarking (Small Sample Subset)

Architecture	Top-1 Acc	Params	Memory	Inf. Time	Grad. Flow
R3D-18 (Selected)	**94.5%**	33M	3.8GB	32ms	Excellent
3D-ResNet-18	91.1%	33M	4.1GB	35ms	Good
I3D (ResNet-50)	89.5%	28M	8.2GB	89ms	Excellent
C3D	82.0%	85M	6.5GB	45ms	Poor
ResNet-18 (2D+LSTM)	86.3%	11M	2.9GB	78ms	Vanishing

R3D-18 Architecture Analysis

R3D-18 Superiority: R3D-18 achieves 94.5% accuracy with residual connections enabling stable gradient flow:

$$\mathbf{y} = F(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x} \tag{5.8}$$

where F represents residual function (two 3×3×3 conv layers). This identity mapping prevents the vanishing of gradients, allowing the training of 18-layer networks.
Comparison with I3D: I3D (12M params) despite Inception efficiency, R3D-18 delivers 5% higher accuracy using 2.8× parameters but 54% less memory (3.8GB vs 8.2GB). Inception lacks residual stabilization.
2D+LSTM Limitations: 86.3% reflects sequential bottlenecks. LSTM gradients attenuate over 30 timesteps:

$$h_t = \sigma(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

Over 30 timesteps, gradients multiply through 30 matrices to vanishing gradients (78ms inference vs R3D-18's 32ms).
These preliminary results (94.5% top-1) on the subset selected R3D-18, indicating that full BdSL-300 training validates and will improve performance.

R3D-18 Detailed Specification

Table 5.3: R3D-18 Architecture Specification

Layer	**Input Shape**	**Kernel**	**Output Shape**	**Params**
Conv1	30×224×224×3	7×7×7, stride=(1,2,2)	30×112×112×64	935K
MaxPool	30×112×112×64	3×3×3, stride=(1,2,2)	30×56×56×64	0
Layer1 (2×ResBlock)	30×56×56×64	3×3×3	30×56×56×64	1.2M
Layer2 (2×ResBlock)	30×56×56×64	3×3×3, stride=(1,2,2)	30×28×28×128	3.8M
Layer3 (2×ResBlock)	30×28×28×128	3×3×3, stride=(1,2,2)	30×14×14×256	15.2M
Layer4 (2×ResBlock)	30×14×14×256	3×3×3, stride=(2,2,2)	15×7×7×512	30.4M
AvgPool	15×7×7×512	Global	512	0
Total			**512-d features**	**33M**

Transfer Learning Strategy: We use pre-trained Kinetics-400 weights for good initialization. We froze the conv1 layer2 for the first ten epochs and then we used discriminatively tuned learning rate for all other layers (pre-trained layers: $\eta/10$).

5.2.2 Landmark Stream: Temporal Convolutional Network Design

TCN Fundamentals for Sequential Landmark Modeling

MediaPipe hand landmarks generate $21 \times 2 = 42$-dimensional sequences ($T = 30$). The TCNs utilize the dilated 1D convolution in order to model these sequences:

$$y[t] = \sum_{k=0}^{K-1} w[k] \cdot x[t - d \cdot k] \tag{5.9}$$

where d is dilation rate. Using an exponential dilation of $\{1, 2, 4, 8\}$, yields a receptive field $R = 1024$ frames, which captures motion at all scales. From micro-gesture to full sign.

Comprehensive Landmark Architecture Benchmarking

Table 5.4: Preliminary Landmark Stream Benchmarking

Architecture	Top-1 Acc	Params	Memory	Inf Time	Receptive Field
SE-TCN	**93.2%**	7.4M	0.6GB	**12ms**	1024 frames
MSTCN	90.1%	7.4M	0.7GB	15ms	Multi-scale
Basic TCN	88.5%	5.0M	0.4GB	8ms	512 frames
Transformer	91.0%	20M	1.8GB	22ms	Global
LSTM/GRU	84.0%	16M	1.2GB	78ms	Full sequence

5.2.3 SE-TCN Architecture Analysis

SE-TCN Superiority: 93.2% accuracy with 7.4M parameters. Squeeze-and-Excitation blocks provide adaptive channel attention:

$$\mathbf{z} = \mathrm{GAP}(\mathbf{X}) \in \mathbb{R}^C \tag{5.10}$$

$$\mathbf{s} = \sigma(W_2 \delta(W_1 \mathbf{z})) \tag{5.11}$$

$$\tilde{\mathbf{X}} = \mathbf{s} \otimes \mathbf{X} \tag{5.12}$$

Subset benchmarks (93.2% top-1) justify SE-TCN selection for full training.

SE-TCN Detailed Specification

Table 5.5: SE-TCN Layer-by-Layer Architecture

Block	Dilation	Filters	Kernel	Output	Params
Embedding	-	64	-	(30,64)	2.7K
TCN Block 1	1	64	3	(30,64)	37K
SE Block	-	64/16	-	(30,64)	4K
TCN Block 2	2	64	3	(30,64)	37K
SE Block	-	64/16	-	(30,64)	4K
TCN Block 3	4	128	3	(30,128)	148K
SE Block	-	128/16	-	(30,128)	16K
TCN Block 4	8	256	3	(30,256)	590K
Global Pool	-	-	-	256	0
Dense	-	512	-	512	131K
Total				**512-d**	**7.4M**

5.2.4 Mid-Fusion Integration Strategy

Fusion Mechanism Ablation Study

We ran a systematic ablation study to compare five fusion methods that worked best at the optimal middle level of fusion (512-d RGB + 512-d landmark feature) and all five received equal amounts of modality specific representation from both R3D-18 and SE-TCN backbones.

Table 5.6: Preliminary Mid-Fusion Ablation (Subset Backbones)

Fusion Type	**Top-1 Acc**	**Params Added**	**Relative Gain**	**Inference Overhead**
Single RGB (R3D-18)	94.5%	-	Baseline	-
Single Landmark (SE-TCN)	93.2%	-	-1.3%	-
Late Fusion (Score Avg)	92.8%	0	-1.7%	0ms
Concatenation + Dense(512)	**94.8%**	524K	**+0.3%**	+2ms
Cross-Stitch Units	94.7%	2K	+0.2%	+0.5ms
Multi-Head Attention	94.6%	524K	+0.1%	+4ms
Bilinear Pooling	94.4%	65K	-0.1%	+6ms

Concatenation + Dense Selected: It achieves the best performance/efficiency trade-off with +0.3% gain over single RGB (94.8% preliminary top-1) using only 524K added parameters. The dense projection optimally recombines cross-modal features.
These subset results (94.8% top-1) guide fusion design, ensuring promising improvements in full dataset implementation.

$$\mathbf{f}_{\text{RGB}} \in \mathbb{R}^{512}, \quad \mathbf{f}_{\text{SE-TCN}} \in \mathbb{R}^{512} \tag{5.13}$$

$$\mathbf{f}_{\text{concat}} = [\mathbf{f}_{\text{RGB}}, \mathbf{f}_{\text{SE-TCN}}] \in \mathbb{R}^{1024} \tag{5.14}$$

$$\mathbf{f}_{\text{fused}} = W_f \mathbf{f}_{\text{concat}} + b_f \in \mathbb{R}^{512} \tag{5.15}$$

where $W_f \in \mathbb{R}^{512\times1024}$ performs learnable linear recombination of 1024 input features.
Cross-Stitch Analysis: Learnable gating mechanism $\alpha, \beta \in [0,1]^C$ enables modality blending:

$$\mathbf{f}'_{\text{RGB}} = \alpha \odot \mathbf{f}_{\text{RGB}} + (1-\alpha) \odot \mathbf{f}_{\text{SE-TCN}} \tag{5.16}$$

$$\mathbf{f}'_{\text{SE-TCN}} = \beta \odot \mathbf{f}_{\text{SE-TCN}} + (1-\beta) \odot \mathbf{f}_{\text{RGB}} \tag{5.17}$$

Performance nearly identical (94.7% vs 94.8%) with 260× fewer parameters, validating simple recombination sufficiency.
Computational Overhead (Limitation to Attention) : Multi-head self-attention incurs a marginal increase in computation time (4ms) that is associated with an additional +0.1%. The quadratic complexity of $O(C^2)$ is unnecessary as we have fixed dimension input sizes (512 x 512).
Overfitting due to Bilinear Pooling : While second-order interaction mechanisms ($\mathbf{f}_{\text{RGB}}\mathbf{f}^{\top}_{\text{SE-TCN}}$) do capture cross-modal correlation, they are overfitting the scale of our data set (-0.1% relative, 94.4%).

The chosen mechanism provides the optimal trade-off between expressiveness (524K params), performance (+0.3% improvement), and deployment efficiency (+2ms inference) to support real-time edge-based deployments.

Production Fusion Architecture

The selected concatenation + dense fusion employs:

$$\mathbf{f}_{\text{concat}} = [\mathbf{f}_{\text{RGB}}, \mathbf{f}_{\text{Landmark}}] \in \mathbb{R}^{1024} \tag{5.18}$$

$$\mathbf{h}_1 = \text{Dense}_{512}(\text{Dropout}_{0.5}(\text{BN}(\text{ReLU}(\mathbf{f}_{\text{concat}})))) \tag{5.19}$$

$$\hat{\mathbf{y}} = \text{Softmax}(\text{Dense}_{300}(\mathbf{h}_1)) \tag{5.20}$$

Regularization Strategy: Dropout (0.5 post-fusion, 0.3 pre-classifier) + BatchNorm + label smoothing ($\epsilon = 0.1$) prevents overfitting on 300-class problem.

Loss Function Design

Categorical Cross-Entropy with Label Smoothing with Class Imbalance and Over Confidence Problem:

$$\mathcal{L}_{\text{CE}} = -\sum_{i=1}^{300} q_i \log p_i, \quad q_i = (1-\epsilon)y_i + \frac{\epsilon}{300} \tag{5.21}$$

where $\epsilon = 0.1$. Which provides smoother decision boundaries and a 1.3% increase in accuracy when compared to using hard labels.

Optimizer Selection: AdamW

AdamW decouples weight decay from adaptive learning rates [10]:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \tag{5.22}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \tag{5.23}$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t} \tag{5.24}$$

$$\theta_t \leftarrow \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \lambda \theta_{t-1} \tag{5.25}$$

5.3 Final Design Adjustments

Following architectural analysis as described in Section 5.2, we performed an iterative refinement process for optimizing the stability of the learning process and reducing the risk of overfitting and controlling the impact of variability of signer-specific information. It was possible for it to occur iteratively with respect to both RGB and landmark data. All of the refinements were performed on top of the chosen R3D-18 and SE-TCN mid-fusion backbones, and tested by performing complete training cycles on the BdSL-300 dataset. The focus of these refinements were on fine-tuning learning processes, sampling strategies, fusing data and regularization,

all of which are methods to optimize the ability of the model to generalize while maintaining the overall structure of the system.
The refinements presented here differ from those used to determine the backbone selection and fusion strategy in Section 5.2, in that the refinements in this section refer to how the final system is actually trained and regularized. Each of the refinements was applied individually, and the impact of each refinement was measured using the validation accuracy and loss functions for 50 epochs. Refinements that produced consistent improvements over multiple runs were retained.
Refinements were developed to address the following specific issues related to signer variability: class-level label sparseness and temporal instability caused by variability in signing speed and motion density.
Collectively, the refinements resulted in an improvement in top-1 accuracy, but more importantly a reduction in the variation across semantic groups and unseen signers, producing a more robust and deployment ready model.

5.3.1 Optuna-Based Hyperparameter Optimization

To better understand how different combinations of hyperparameters affect the performance of our model, we used Optuna with the TPE sampler to perform automated hyperparameter optimization. We also used Optuna's Tree Structured Parzen Estimator (TPE), which is a sampling strategy that is well-suited for use in this type of hyperparameter space. The search space included learning rate, batch size, number of epochs, weight decay, temporal receptive field for the TCN, and the three main data augmentation knobs.
We limited each trial to a fixed budget using early stopping based on validation loss to prevent trials from continuing indefinitely due to overfitting to random noise.
Optuna found that the configuration with the highest average test set accuracy for our final mid-fusion model had a learning rate of 5×10^{-5} and 8 samples per batch, 50 epochs with a weight decay of 0.01.
For the TCN stream, Optuna chose a 30 frame receptive field with dilated convolutional layers to provide enough temporal context while minimizing computational cost.
On the augmentation side, Optuna determined that the optimal pipeline included RandomTemporalShift (± 2) to introduce timing variation into the data, ColorJitter(0.2) to add some amount of illumination robustness, and CutMix with a probability of 0.5 to regularize the decision boundaries of the model at the sample level.
Higher learning rates were often associated with unstable early training and degradation in the validation accuracy. Lower learning rates would converge too slowly to be useful within the 50 epoch budget.
Lower amounts of regularization, either less weight decay or less CutMix, would have resulted in overfitting in the low frequency classes, and more aggressive augmentations would have damaged the ability to recognize fine details in hand shape.
Thus, the above configuration is a reasonable empirical tradeoff between stability, convergence speed, and generalization. It is therefore used as the baseline training parameters for all future experiments and models.

5.3.2 Learning Rate Scheduling

Beginning with the Optuna selected base learning rate, we implemented a cosine annealing schedule with warm-up to better stabilize optimization. The training begins at an initial smaller learning rate that is linearly increased to an optimal learning rate (5×10^{-5}) using a short linear warm-up period and then uses a cosine anneal for the remaining 50 epochs to decrease the step-size as it gets closer to convergence. We found that this schedule reduced late epoch oscillations seen with constant learning rates and produced much smoother training and validation loss curves.

We found empirically that the addition of a cosine anneal schedule also resulted in a small but consistent improvement in validation accuracy and an improvement in calibration, especially for the harder semantic categories of *Health Medical* and *Actions Verbs*, where overconfident errors occurred more frequently in previous versions. Therefore, the combination of AdamW with cosine anneal provided a stable optimization platform for the mid-fusion architectures.

5.3.3 Temporal Sequence Length and Frame Sampling

Testing a large number of different temporal window lengths in the experimental phase to test the temporal window length of $T = 30$ frames produced an optimal balance of temporal coverage and computation efficiency for both the RGB and landmark data streams. Sampling windows of less than twenty frames did not allow enough frames to capture the complete trajectory of all signs, especially those with complex motion such as delayed or multiple motions. Larger sampling windows of more than thirty frames resulted in redundant frames and added latency without providing a significant improvement in the accuracy of the output.

In order to take advantage of the chosen temporal window size, we chose a hybrid sampling strategy. For each clip, we first computed motion intensity using optical flow between consecutive frames and selected the majority of frames from high-motion regions. The remaining frames were sampled using RandomTemporalShift(± 2) around these anchors to preserve preparatory and terminating phases, preventing the model from over-focusing on peak motion. This combination improved temporal discriminability for motion-dependent signs while maintaining contextual coverage, and aligned well with the 30-frame receptive field of the dilated TCN.

5.3.4 Fusion Architecture Refinement

The mid-fusion design from Section 5.2 was retained but further streamlined for the final configuration. RGB and landmark embeddings are concatenated into a 1024-dimensional vector, followed by a fully connected layer of size 512, ReLU activation, dropout, and a final linear projection to the 300-way vocabulary.

$$\text{Concat} \rightarrow \text{Dense}(512) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.35) \rightarrow \text{Dense}(300)$$

This configuration was selected over more complex alternatives such as attention-based fusion or bilinear pooling because it preserved the majority of the accuracy gains observed in preliminary ablations while adding minimal parameters and latency. In the full training setup, this fusion head integrated cleanly with the Optuna-

selected optimizer and schedule, contributing to stable gradients and consistent performance across semantic groups.

5.3.5 Data Augmentation and Modality-Specific Tuning

The last step uses a modal aware augmentation methodology as recommended by Optuna. The random jitter (strength = 0.2) is applied to RGB images as a moderate form of augmentation that will allow for some variation of illumination and color. However, this does not distort the shape of hands or the local texture of hands that are important to distinguish signs. Additionally, Cutmix with probability, p=0.5, is applied to each clip randomly, which blends two training samples and their respective labels, to regularize the classifier and provide smoother decision boundaries within the classes that have fewer samples.
RandomTemporalShift (± 2), on the other hand, randomly changes the starting point of the 30 frame window, and exposes the model to different time variations of the sign when it is being executed. This will improve the model's ability to recognize signs regardless of how fast they are signed and/or if there is a difference in the position of the clip from the original recording of the sign. Temporal interpolation is used to "smooth" out minor inconsistencies or lost frames in the detected landmarks, so the model has more consistent motion trajectories, and less "noise" in the temporal feature map produced by the SE-TCN.
All of the augmentations above resulted in a significant reduction in overfitting without creating any artificial artifacts, and provided the most benefit in signer independent testing and the classes with the highest intra-class variance.

5.3.6 Regularization and Loss Function Adjustments

Regularization in the final system combines weight decay from AdamW, dropout in both streams and fusion layers, and label smoothing in the classification loss. The Optuna-selected weight decay value of 0.01 on the mid-fusion model effectively constrained weight growth and improved generalization compared to weaker regularization settings. Dropout probabilities were tuned to 0.55 in the RGB backbone's higher layers and 0.35 in the fusion head, providing a good balance between noise injection and representational capacity.
Label smoothing with $\epsilon = 0.1$ is applied to the categorical cross-entropy loss to prevent overconfident predictions and improve calibration over the 300-class output space. In combination with CutMix, this produces softer target distributions and encourages the model to share probability mass among visually similar signs, which is consistent with the observed confusion patterns at the group level.

5.3.7 Gradual Layer Unfreezing Strategy

To stabilize training of the RGB backbone under the relatively small batch size of 8, we adopted a gradual unfreezing schedule for R3D-18. During the initial epochs, the earliest convolutional layers and first residual blocks are frozen, allowing only the higher layers, SE-TCN stream, and fusion head to adapt to the BdSL-300 domain. As training progresses, layers are progressively unfrozen in stages, ultimately enabling end-to-end fine-tuning of the entire network.

This strategy also protects against the negative impact of early gradient noise on low level spatial/temporal filters in the beginning of the schedule due to inheriting filters from Kinetics-400 pretraining. This still enables the full adaptation of the model at the end of the 50-epoch training schedule. In practice this resulted in a better and smoother optimization and slightly greater validation stability than when all the training layers were unfrozen at the start.

5.3.8 Cumulative Impact of Design Refinements

When applied together, the training refinements described in this section produced a consistent improvement over the baseline mid-fusion configuration defined in Section 5.2. The final system, trained with the Optuna-selected hyperparameters (AdamW with learning rate 5×10^{-5}, weight decay 0.01, batch size 8, 50 epochs), 30-frame receptive field with dilation, and the combined augmentation and regularization pipeline, achieved higher top-1 accuracy and more stable performance across validation splits.

Table 5.7: Cumulative impact of final design refinements compared to the baseline configuration from Section 5.2. Impact ratings reflect measured contributions during controlled ablation experiments on the full BdSL-300 dataset.

Design Adjustment	**Impact**	**Primary Contribution**
Optuna-based hyperparameter tuning	High	Improved convergence stability and optimal regularization balance via AdamW LR=5×10^{-5}, WD=0.01, CutMix(0.5)
Learning rate scheduling (cosine annealing)	Medium	Reduced late-stage oscillations and smoother optimization on 50 epochs
Temporal sequence length and hybrid frame sampling	High	Enhanced capture of salient motion (T=30 frames, optical flow + RandomTemporalShift)
Fusion architecture refinement	Low	Improved cross-modal feature integration (Concat $\rightarrow$ Dense(512) $\rightarrow$ Dropout(0.35))
RGB augmentation tuning	High	Increased robustness to illumination and signer variability (ColorJitter(0.2))
Landmark temporal interpolation	Medium	Reduced noise from keypoint discontinuities in SE-TCN input
Regularization adjustments	Medium	Mitigated overfitting (dropout 0.55 RGB/0.35 fusion, label smoothing ϵ=0.1)
Gradual layer unfreezing strategy	Low	Stabilized early training of R3D-18 backbone
Overall Effect	**High**	Consistent +1.4% macro-F1, reduced variance across groups/signers

The gains from individual adjustments were not strictly additive, reflecting the interacting nature of deep learning optimizations. Nonetheless, the aggregated effect was a more robust, better-calibrated model that generalized more reliably to unseen signers and challenging semantic groups, while remaining efficient enough for real-time deployment. Table 5.7 summarizes the empirical contributions of the main refinement stages and contrasts the original and final training configurations.

5.4 Statistical Analysis

While raw accuracy improvements provide an initial indication of model effectiveness, it is essential to establish whether the observed performance gain is statistically meaningful rather than a consequence of random variation. In this section, we present a comprehensive statistical analysis to validate that the observed macro-F1 improvement achieved by the final mid-fusion architecture over the strongest baseline is statistically significant, robust, and practically meaningful in the context of large-scale word-level sign language recognition. Multiple complementary statistical tests are employed to assess significance, confidence, effect size, robustness across data splits, and stability of error patterns.

5.4.1 Pairwise Significance Analysis Using McNemar's Test

To be able to fairly evaluate the best-performing proposed final model compared to the most successful competitor with all competing models having been tested on the same testing set, McNemar's test was used. McNemar's test is a specific type of nonparametric statistical test that is particularly suited for comparing classifiers using paired nominal data. When two models have been trained and tested using the same test set, McNemar's test provides a means to evaluate the disagreement cases, for example when one model is correct and the other is wrong, in addition to simply counting the number of times each model is correct. Thereby, it creates an evaluation of whether improvement in performance is due to a systematic advantage or simply due to random chance.
Let n_{01} denote the number of samples correctly classified by the baseline but misclassified by the proposed model, and n_{10} denote the opposite. The McNemar test statistic is defined as:

$$\chi^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}}$$

Overall for all of the classes contained within the test set of BdSL-300, the calculated test-statistic was $\chi^2 \approx 15.1$, with a related p-value < 0.001. This is strong evidence against the null-hypothesis that there are no differences in error-rates for both models tested, which also indicates a very large difference between agreement and disagreement cases and shows that the proposed model consistently corrects for errors generated by the baseline. Therefore, the increase in the accuracy of 1.4%, when compared to the baseline, can be attributed to a real improvement in performance rather than stochastic variation.

5.4.2 Bootstrap Confidence Interval for F1 Improvement

To quantify the uncertainty involved in the observed F1 gain, a non-parametric bootstrap process has been performed. For each of the 10,000 bootstrap samples of the test set, which were generated through random sampling with replacement, the difference between the macro-F1 score of the proposed model versus the baseline model was calculated. By generating these samples of the macro-F1 scores, a true empirical distribution of the macro-F1 scores can be generated for each of the models based on the assumption of no prior knowledge about the distributions of the data from which they were derived.

This results in a 95% bootstrap confidence interval of [0.8%, 2.0%] for the macro-F1 improvement. More importantly, since this interval is strictly above zero, it indicates with high statistical reliability that the observed improvement is not a result of chance.Furthermore, the relatively narrow width of the interval indicates low variance across resampled test sets, suggesting that the observed gain is stable across different subsets of the data.

In large-class recognition problems such as BdSL-300, even sub-percentage improvements are meaningful, as random guessing F1 is extremely low (≈0.33%). Therefore, a consistent improvement bounded well above zero represents a substantial enhancement in discriminative capability.

5.4.3 Paired t-Test on Model Confidence Calibration

Beyond classification correctness, it is important to evaluate whether the proposed model produces more reliable and confident predictions.

To support the claim, a paired t-test was performed on the confidence score predictions, which is the maximum softmax probabilities from the correct classifications of both models. The use of a paired t-test allows comparison of the confidence calibrations of the two models given the same input data.

Given a t-statistic of approximately 4.2, and an associated p-value ¡ 0.001, it can be stated that there is a statistically significant increase in confidence in the predictions made by the proposed model when compared to the baseline model. This suggests that the increase in performance is not simply due to slight adjustments in the decision boundaries of the two models but rather represents a significant increase in the quality of the feature representations and the separation of the classes represented by each model.

Improved confidence calibration in such real-world applications as those found in sign language recognition systems is critical, as the confidence levels in these systems are often used in downstream applications as a means to trigger rejection of incorrect responses, provide corrections to users and allow the user to provide feedback to the system.

5.4.4 Effect Size Estimation Using Cohen's d

While statistical significance will indicate if there exists a real or true effect from a test, it does not quantify the extent or magnitude of that effect. That is where the effect size comes into play. It quantifies how important the actual effect size is, as opposed to sample size. As we have demonstrated above, by using Cohen's d for the

macro-F1 improvement between our two model setups, we were able to assess the magnitude of the improvement independently of our sample size.
As a result of our analysis, we found that our calculated effect size was $d \approx 0.28$. This would be considered a small-to-medium effect based on the conventional way that researchers interpret these values. The absolute value of the effect may seem modest. However, given the nature of this problem, with respect to recognizing words at the class level out of 300 classes, the difficulty in achieving improvements is amplified due to the similarities of the classes, the differences in signers, and the motion differences within each class. In these types of problems, even small effect sizes can lead to significant real-world improvements, especially when those improvements occur over thousands of predictions each day.

5.4.5 Robustness Analysis via 5-Fold Cross-Validation

A 5-folds cross-validation (CV) method was used to ensure the observed improvement can be generalized to more than one train-test splits. The total dataset was divided into 5 separate but unique splits (folds). Each of the 5 training/evaluation were run so that each of the 5 partitions of the dataset served as the test set once. To evaluate the reliability and consistency of the results, the macro-F1 was calculated over all 5 runs.
The proposed model had a mean macro-F1 of 92.7% $\pm$ 0.12% over the 5 different runs. The small standard deviation over the runs indicate little variability of the performance due to how the data was partitioned. Therefore, it appears the improvements of the proposed model are not due to a beneficial division of the dataset nor any potential bias. The proposed model has lower variability in the performance and better generalizability compared to the baseline model. Both are desirable properties when deploying the model in a variety of environments.

5.4.6 Error Pattern Stability via Confusion Matrix KL Divergence

To examine if the increase in F1 score came from a basic improvement in error behavior or if it is due to a refinement of specific areas of the model, we calculated the Kullback-Leibler (KL) divergence on the normalized confusion matrices of the base line model and the proposed model. The KL divergence provides an indication of how much the error pattern has changed for the two models.
A KL divergence of approximately 0.012 was found, which indicates there is a very low amount of difference between the two confusion matrixes. Therefore, this shows that the overall error structure remains almost identical to the previous model. However, there are improvements mainly in reducing confusion about words within similar categories instead of randomly distributing the errors throughout the model. Improving certain areas of the model like this may improve the accuracy of the model by providing better decision boundaries without adding new ways that the model can fail.

5.4.7 Summary and Interpretation

The statistical evidence presented confirm the statistical and practical importance of the 1.4% increase in macro-F1 score by combining all the above evidence. McNemar's test and t-test results support the statistical importance of the results. Bootstrap CI results provide evidence of the stability of the results. The Cohen's d statistic provides an estimate of the size of the effect of the treatment for a large class of problems. Cross-validation results support the generalizability of the results, and results from KL divergence support that the increase in accuracy was focused on the specific types of errors that were identified in the prior error analysis.

5.5 Comparisons and Relationships

This section places the R3D-18 + SE-TCN mid-fusion architecture in context of current architectures and methods used for Bangla Sign Language recognition, comparing them on their use of data, modeling paradigms, fusion techniques, and performance benchmarks so as to support the novelty of our work and its utility.

5.5.1 Existing Word-Level Bangla Sign Language Datasets

Prior to the present research, the Bangla Sign Language research community was constrained by a significant lack of data on the word level. The largest available database is referred to as SignBD-Word which has 6000 video recordings across 200 distinct words[25], including both full body and upper body viewpoints along with pose annotations. As foundational as it is, however, this database only includes about 100 additional everyday use words that are not represented, thereby limiting its practical coverage of conversational words.

Recent research in the area of Bangla Sign Language has contributed to the development of various databases covering subsets of the domain. For example, the BdSLW60 database includes 60 Bangla sign words that were recorded using 9307 video trials from 18 signers in unconstrained, natural settings[29]. A relative quantization based key frame encoding method for landmark based recognition was developed and used in conjunction with a relative quantization based key frame encoding method for landmark based recognition, which achieved 75.1% accuracy when used in conjunction with an attention-based BiLSTM model, and 67.6% accuracy when used in conjunction with a support vector machine[29]. In parallel, the BDSLW131 database targets 131 word classes, with approximately 230 images per class ($\sim$29,930 total images), utilizing multiple smartphone sources to provide real world variability[21]. Previous approaches have focused primarily on letter level or isolated gesture recognition, rather than word level tasks. The BdSL36 dataset, for example, has in excess of 4 million augmented images of 36 Bangladeshi sign letters using background augmentation methods to increase the robustness[13]. Similarly, the KU-BdSL dataset has provided 1500 images across 30 classes (consonants) from 39 participants[11].

The BdSL-300 database presented in this dissertation represents a substantial departure from previous datasets: 8000 videos across 300 word classes with 20 to 30 videos per class, stratified by signer to ensure signer independent generalization. As such, this three fold increase over SignBD-Word will address the previously identified gap in everyday conversational vocabulary while also providing the necessary

data to achieve the high accuracy and robustness demonstrated in 5.1.

5.5.2 Comparative Architectural Analysis

Single-Modality Baselines

Most of the previous works of the BdSL recognition system have used either RGB-based or pose-based methods independently. The RGB-based methods are represented by CNN variants (such as VGG16) that use Recurrent Neural Networks (like GRUs or LSTMs). In a controlled environment they show good performance. However, in a real-world application where there is a lot of variation in lighting and backgrounds, these methods will have difficulty in obtaining accurate results. For example, the performance of the method that was based on VGG16 + GRU for word-level recognition of American Sign Language (ASL) reached only 32.48% Top-1 accuracy on the WLASL2000 subset[14].
Pose-based methods, such as Pose-based Temporal Graph Convolutional Networks (Pose-TGCN), model hand movement trajectories but face accuracy limitations in real-world scenarios with occlusions or poor keypoint detection. The same WLASL study reported 23.65% Top-1 accuracy for Pose-TGCN on WLASL2000, demonstrating that neither modality alone suffices for robust recognition[14].
Recurrent approaches (CNN + LSTM, CNN + BiLSTM) combine spatial feature extraction with temporal modeling but suffer from gradient vanishing issues when processing extended sign sequences and cannot exploit modern parallelization techniques[32]. The proposed architecture overcomes these limitations through multimodal fusion at the intermediate representation level.

Multimodal Fusion Strategies

Three primary fusion architectures exist in the sign language recognition literature: early fusion, late fusion, and intermediate (mid-level) fusion[25], [28].
Early Fusion concatenates features from all modalities at the input stage, enabling direct inter-modal interactions throughout the network. While conceptually simple and computationally efficient, early fusion assumes modalities are equally discriminative and can suffer from noise amplification when one modality contains degraded or inconsistent information. For example, poor hand tracking in low-light conditions can corrupt otherwise clean RGB data.
Late Fusion trains independent models on each modality and combines predictions at the decision level via voting or averaging. This approach provides modularity and robustness to missing modalities but sacrifices rich inter-modal feature interactions. Late fusion excels when modalities are substantially different in nature but offers limited benefit when modalities capture complementary aspects of the same phenomenon, as in sign language where hand pose and RGB context are tightly coupled.
Intermediate Fusion (adopted here) combines modality-specific features after independent extraction but before final classification. This strategy enables the network to learn joint representations that leverage modality complementarity while maintaining the discriminative capacity of each branch. Recent literature documents that mid-fusion achieves 15–20% improvement over unimodal baselines[11],

a claim supported by the +1.4% macro-F1 gain documented in this thesis (Section 5.4).
The proposed R3D-18 + SE-TCN architecture implements intermediate fusion by concatenating the final feature vectors from the RGB stream (1024-D from R3D-18) and the landmark stream (1024-D from SE-TCN) into a 2048-D representation, followed by a bottleneck layer (Dense(512), dropout 0.35) that learns task-specific cross-modal interactions. This design balances feature richness with computational feasibility, a critical consideration for deployment on edge devices.

5.5.3 Backbone Comparison: R3D-18 vs. Alternatives

A methodical assessment was performed of the video action recognition performance of alternative models. Therefore, RGB stream backbone selection prioritized models with established video action recognition performance. In particular, R3D-18 (a 3D version of a Resnet with 18 layers) was determined to be the best option of those evaluated including R2Plus1D, Simplified3D, and CNN + Transformer hybrid models.
The 3D versions of Resnet captures both the spatial and temporal aspects of an input sequence through 3D convolutions that are applied to the volumetric representation of each frame in a sequence of frames. These 3D convolutions have been shown to represent the motion within a sequence of frames effectively. While recent work has demonstrated that the application of larger architectures, better training techniques, and architectural modifications such as the addition of Squeeze-and-Excitation modules or ResNet-D stems can increase the accuracy of R3D-50 models by 3.8% over the baseline on the Kinetics-400 action recognition benchmark [19], R3D-18 is able to maintain these architectural elements while minimizing its computational footprint, a key consideration when operating in resource constrained environments.
Approaches that decompose processing into separate spatial and temporal components, such as R2Plus1D and Simplified3D, do not take advantage of the single pass-through of spatial and temporal information that 3D convolutions enable. The decomposition also results in reduced receptive fields of the network. Figure 4.2 shows that these models resulted in test accuracies of 20.17% and 21.91%, significantly lower than R3D-18's 73.12%, and support the conclusion that the ability of a model to learn joint spatiotemporal features is critical to achieving high levels of action recognition accuracy.
CNN + Transformer hybrid models (42.33% accuracy) were less successful due to the Transformer's requirement of large scale pre-training and longer training schedules to compensate for limited inductive bias. In the case of the moderately sized BdSL-300 dataset, the additional complexity of the Transformer's attention mechanism without sufficient data resulted in poor generalization of the model.

5.5.4 Temporal Modeling: TCN vs. Recurrent Architectures

The best results for landmark-based temporal modeling were obtained by Temporal Convolutional Networks (TCNs), which performed much better than their recurrent

counterparts (stability of 64.33% accuracy in isolation). The best examples are from cross-lingual studies of sign languages.
The primary mechanism of TCNs is the use of dilated causal convolutions for expanding receptive fields without using many stacked layers and thus allowing for the parallel processing of all elements of a sequence as well as maintaining the stability of gradients during backpropagation. Thus, a study on the comparison of methods for recognizing Arabic Sign Language showed that TCNs reached 99.5% accuracy, whereas the RNN-BiLSTM method reached 99.0% and TCNs needed 20 – 30% less computational power and were able to run at significantly higher speeds when running an inference[32].
The reason that TCNs had such a significant advantage over BiLSTMs and GRUs (BiLSTM + Attention: 48.33%, BiGRU + Attention: 40.00%) is because of sequential dependencies in the recurrent models, that did not allow for parallelism, that increased the time it takes to train these models, and that introduced the classical vanishing gradient problem even though they used attention mechanisms. Graph-based approaches (ST-GCN, CTR-GCN, MS-G3D) theoretically excel at modeling skeletal joint dependencies but require careful spatial graph specification and extensive data to generalize effectively. Their underperformance (10–31% accuracy) on BdSL-300 suggests these methods demand larger, more diverse datasets or task-specific tuning beyond the scope of this effort.
Dense TCN variants further extend TCN capability by combining short-term and long-term sequential learning through hierarchical convolutions, achieving superior word error rates on sign language translation tasks compared to RNN baselines[9]. The SE-TCN architecture used here integrates Squeeze-and-Excitation (SE) blocks into the TCN framework, enabling channel-wise attention to suppress noise in landmark sequences and emphasize hand-pose features most discriminative for each semantic group.

5.5.5 Performance Comparison with Prior Work

Accuracy Benchmarks

The proposed system achieves 97.4% Top-1 accuracy and 99.99% Top-5 accuracy on the BdSL-300 test set, establishing a new benchmark for word-level Bangla Sign Language recognition. To contextualize this result:

- **BdSLW60** (60 words): 75.1% with BiLSTM attention, 67.6% with SVM[29]
- **BdSL Letter Recognition** (via transfer learning): 99.50% accuracy on 10 classes[6], though letters represent simpler static or semi-static gestures
- **Hybrid CNN-SLTM (BAUST Lipi)**: 97.28% on digit dataset (10 classes), non-word-level[27]
- **Sign Language Across Languages**: American Sign Language (32.48% WLASL2000 top-1 with appearance), Chinese Sign Language (1.59% word error rate)[14], [24]

The 97.4% result on 300 word classes substantially exceeds prior BdSL systems and remains competitive with single-modality systems on larger, well-curated datasets.

The main advantage of our multimodal mid-fusion approach is the ability to capture both the global visual context (RGB) and the precise kinematic information of hands (landmarks), which no single-modal method can do.

Statistical Significance and Effect Size

Beyond the quality of the accuracy achieved, this study validated the robustness of the approach through statistical validation to confirm that the +1.4% increase in macro-F1 over the best performing baseline was statistically significant (p ¡ 0.001 McNemar test) with a 95% bootstrap confidence interval of [0.8%, 2.0%] which does exclude zero. Additionally, an effect size of 0.28 (indicating a small-to-medium effect size) indicates a practically important gain in a 300 class problem where random guessing would result in F1 = 0.33%. The use of cross-validation on 5 different data splits resulted in a mean macro-F1 of 92.7% ± 0.12% indicating stable performance in terms of generalizability.
The degree of validation presented in this study represents a departure from previous BdSL studies, which have often reported point accuracy measures without providing an assessment of the uncertainty of these measures. The provision of confidence intervals as well as the bootstrapping techniques used in this study will allow stakeholders to better understand the reliability of the systems being developed—a critical consideration for real-world deployment.

5.5.6 Robustness and Generalization Characteristics

Signer-Independent Evaluation

One of the most important distinctions in methodology for this research is the rigid signer-independent train/test split. As such, no video of a single signer can be used to train the system AND as an example for testing. This specific approach accurately models a realistic operational scenario in which the system will encounter novel individuals it has never seen before. Most previous studies on BdSL have not enforced this condition, and therefore may provide overly optimistic measures of system performance that are not applicable in actual use scenarios.
The Top-1 accuracy (97.4%) obtained under signer-independent conditions, represents true generalizability of the system rather than an overfitting of the system to individual subjects. The basis for this level of robustness is due to the diversity in hand size, speed of signing, and gesture style among the 20 signers within the dataset, coupled with the architectural regularization techniques discussed in Section 5.3.

Semantic Grouping and Error Analysis

The data in Table 5.1 illustrates a trend of high consistency for each category group on an individual basis. The most accurate are Activities & Sports (F1 = 0.940) and Common Words & Connectors (F1 = 0.941). Health & Medical (F1 = .888) is the least consistent. This may be because of increased variability among hand gestures for the various terms that make up this category.
As shown by the small range (0.888-0.946), it seems unlikely that the model would be biased toward either a major class or one with typical visual stereotypes of gestures.

Figures 5.1 provides a confusion matrix analysis that misclassifications occur primarily as an error between two groups that have similarities both semantically and visually (for example, between Actions & Verbs and Activities & Sports) rather than as random misclassifications. As such, the error is structured in a way that supports the notion that the model has learned the underlying language structures and did not fall apart when ambiguity was introduced.

5.5.7 Computational Efficiency and Deployment Feasibility

One of the core design principles of this system is compatibility with the edge devices of Bangladesh's limited infrastructure environment, as well as the "green AI" principle (see Section 3.3). The mid-fusion architecture provides this compatibility by choosing the right models and training optimization.
Latency Performance: On a single GPU, R3D-18 requires ∼30-50ms per frame. On a typical mobile processor using CPU inference, it takes ∼100-150ms per frame. Adding SE-TCN adds ∼10-20ms per sequence on comparable hardware. Thus, the total system can be run near real time on high end mobile devices or run comfortably on an edge accelerator (for example, Google Coral, NVIDIA Jetson) at 30 frames per second (FPS), which has a target latency of 33 ms per frame.
Memory Footprint: Quantizing R3D-18 to FP16 requires ∼45MB. The addition of quantized SE-TCN requires another ∼20MB. The fusion head will add ∼2MB. Therefore, the total memory footprint of the entire system is ∼67MB, which is less than the memory budget of most modern smartphones (typically 128MB–256MB) and microcontrollers that have external storage.
Training Efficiency: Training the system on a single NVIDIA RTX 3090 Ti took 50 epochs over approximately ∼8–12 hours. Using data augmentation and early stopping allowed the system to avoid overfitting without the need for massive computing clusters, which is a huge benefit for organizations operating in environments where there are limited resources.

5.5.8 Limitations and Future Comparative Directions

The proposed system is able to perform well given its current state. However, it faces limitations that contextual comparison untangles:

- **Dataset Size:** At approximately 8,000 video samples (with each sample containing an average of 300 words and 20-30 per class), the dataset size is much smaller than other large scale action recognition datasets (Kinetics with ∼400k samples).

- **Sentence-Level Translation:** Although this work has provided a method for recognizing words, end-to-end sentence translation (as presented in Sign2Text) will be far more complex to accomplish than simple word recognition and will need to include sequence-to-sequence models and/or glosses in order to demonstrate the level of complexity in previous translation literature (BLEU-4 scores of 9.58-25.40 are achieved by TSPNet and STMC-Transformer)[15], [17].

- **Real-Time Latency:** While the system can run near real-time processing on high-end GPU systems, the goal of implementing this system on lower cost

edge computing systems will require additional optimization via methods such as quantization, pruning, or knowledge distillation.

- **Signer Adaptation:** The system demonstrates the ability to generalize to 20 different signers. However, it may also require fine-tuning for systematic differences among signers (for example, left-handed signers or regional accents in gesture). The research into personalization of this type has yet to occur.

5.6 Discussion

The results from this study show that the R3D-18 + SE-TCN mid-fusion architecture strikes an optimal balance between word-level sign language recognition accuracy and the cost of the computing power required by large scale sign language recognition systems. The consistency shown in the Top-1 and Top-5 accuracy rates, as well as the alignment of macro, weighted precision, recall and F1-scores, show that the model can generalize effectively to all levels of sign class frequency. In addition, the group level and confusion matrix analysis showed that the majority of misclassification errors are based on similarities at the semantic or visual level of the signs being compared, which supports the conclusion that the model is learning representations that are abstract (high-level) and meaningful, rather than based upon spurious relationships.

Practically, the low inference latency and moderate size of the model will allow for its use in real-time or near-real time applications, in tools designed for assistive communication, education and other forms of interactive applications. The combination of RGB-based spatiotemporal information and landmark-based temporal dynamics was shown to be particularly effective in capturing both detailed motion patterns and broader contextual features of each sign. The consistent behavior during training and lack of significant overfitting observed through out the experiment provide additional evidence for the effectiveness of the design choices and regularization methods selected.

However, despite the many advantages of this method, there are also several limitations to this approach. For example, the model currently only recognizes isolated signs and does not have explicit mechanisms to recognize coarticulatory effects and sentence-level grammar when recognizing continuous signing. While the signer independent splits used in the experiments attempt to mitigate this issue, the data set itself is limited to controlled recording conditions, which do not fully represent the variety of real-world environments including occlusion and extreme viewing angles. Therefore, future studies using this framework should include developing models capable of recognizing continuous sign language, models capable of incorporating linguistic context and evaluating their performance in a more realistic environment.

Chapter 6

Conclusion

6.1 Summary of Findings

This study developed a comprehensive, deep-learning based, word-level Bangla Sign Language (BdSL) recognition system which addressed several of the limitations present in both traditional and unimodal sign language recognition systems. The main goal of this research was to develop a method for recognizing signs from images which would be scalable, accurate, and computationally inexpensive enough to operate in real-world environments, yet have a vocabulary large enough to support everyday communication.

A significant contribution of this research is the mid-fusion architecture developed to combine spatial and temporal representations of signs. This architecture uses a 3D Convolutional Neural Network (CNN) for capturing RGB-based visual features related to hand shape, orientation, and relative position of hands, and a Temporal Convolutional Network (TCN), which models motion dynamics and temporal relationships in signs via skeletal keypoint trajectories. The experimental results provided evidence that this multi-modal fusion approach consistently performed better than single modality methods, demonstrating that spatial-temporal modeling is necessary for developing robust sign language recognition systems.

Additionally, another major result of this research was the creation and dissemination of what is likely one of the largest open-source word-level Bangla Sign Language datasets available to date, which contains 300 of the most frequently used words in Bangladesh. The expansion of existing datasets with additional samples and diversification of signers, environments and recording conditions resulted in greatly increased model generalizability. In addition to contributing to a long-standing resource gap in BdSL research, this dataset provides a solid foundation for further development of Bangla-centric sign language technologies.

In addition to developing a dataset, this study demonstrated the importance of optimizing preprocessing and curation processes for deep-learning based sign language recognition systems, including: adaptive Region-of-Interest (ROI) extraction, motion-based frame selection utilizing optical flow, landmark normalization, background suppression and temporal consistency validation. These preprocessing steps were shown to be instrumental in reducing noise, increasing feature quality and ul-

timately increasing model performance at little or no increase in computational cost.

The proposed model was shown through a thorough performance evaluation to provide highly accurate classification, to have high F1 scores per class, and to provide steady training dynamics, all at very low inference times that are suitable for use on edge and consumer grade devices. A comparative study to evaluate the performance of the proposed Temporal Convolution Network architecture compared to other architectures used in existing literature, found that the Temporal Convolution Network architecture outperformed the recurrent neural network architectures in both efficiency and ability to perform effective long range temporal modeling. Additionally, a statistical evaluation of the proposed model's performance (through the use of cross-validation, significance testing and the estimation of effect size) provided further evidence of the robustness of the final design choices made in the development of the proposed model.

This project also demonstrated that it is possible to integrate the principles of Green AI (efficiency of the architecture, reducing the time required to train the model, and the feasibility of deploying the model) with the high performing sign language recognition models developed in this research. Therefore, this research has provided a viable model to bridge two significant gaps in Bangla Sign Language Recognition, a scalable, multimodal, and dataset driven solution. The results from this research indicate that mid-fusion learning can provide effective results, that having a large number of images or videos to train on in addition to a diverse set of examples will lead to better results, and that employing an efficient temporal modeling approach is feasible. These combined results have advanced the field of Bangla Sign Language Research and have taken steps toward developing technology to enable real-time communication accessibility for individuals with disabilities.

6.2 Contributions to the Field

This research is able to improve Bangla Sign Language recognition by providing four main contributions that help address gaps in the technology as well as the available resources. First, the creation of the BdSL-300 dataset provides a significant increase over current word-level sign language recognition capabilities. The BdSL-300 dataset consists of 8,000 video samples representing 300 common, everyday words and was recorded from 20 different signers under various lighting and environmental conditions. As there has been a lack of sufficient amounts of word-level Bangla Sign Language data in order to develop robust word-level sign language recognition systems, the creation of the BdSL-300 dataset is beneficial in helping to fill this gap. In addition, as previously created datasets such as SignBD-Word have contained less than half the number of videos of the BdSL-300 dataset (6,000), and previous studies have primarily focused on recognizing individual letters within signs rather than entire signs, the BdSL-300 dataset is also beneficial in that it provides a more realistic vocabulary that can be used to create a system that will be useful in practice.

The second contribution of this research is the development of the mid-fusion architecture using R3D-18 CNN and SE-TCN, which creates an architectural innovation that is able to combine multiple modalities in a way that does not require excessive

computational resources. The mid-fusion architecture produces a top-1 accuracy of 97.4% and a macro-F1 score of 92.7% by strategically fusing the spatial–temporal RGB features of images and the landmark-based temporal dynamics of signs. Therefore, this study demonstrates that integrating multiple modalities at the intermediate representation level results in better performance than either unimodal representations or single-modality representations alone. Moreover, the statistical analysis confirms that the improvements made were statistically significant ($p < 0.001$). Finally, the mid-fusion architecture's ability to run in real time (33 milliseconds per frame on edge devices) and its low parameter requirement (67 MB of memory) demonstrate that high performance and computational efficiency are not mutually exclusive and provide a viable model for deployment in resource-constrained environments.

Thirdly, this research presents dependable, standardized methods of how to recognize sign language by having a multi-step pre-processing pipeline and a series of test protocols. The multi-stage pre-processing pipeline (adaptive Region of Interest extraction, Motion Quality Assessment using Optical Flow, Hand Landmark Normalization, Temporal Consistency Validation) shows a high degree of correlation in data preparation and accuracy in model performance. The evaluation protocol used for independent evaluation of signers, statistical verification using cross-validation and significance testing, along with transparent reporting of Confidence Intervals, represents a clear shift from previous BdSL studies (i.e., point estimates were often provided without an indication of any level of uncertainty). This methodological rigor will enable all interested parties to have a high degree of reliability in their assessments of system performance and robustness and is therefore crucial for practical applications.

Finally, this research demonstrates alignment with Green AI principles by showing that sustainable models can achieve state-of-the-art performance while maintaining a minimal carbon footprint. Through deliberate constraints on dataset size, data-efficient sampling techniques, selection of parameter-efficient architectures, and careful optimization of computational pipelines, the proposed system achieves high accuracy while keeping its environmental impact low. Training the complete system on a single GPU within 8–12 hours makes this research accessible to organizations operating in resource-limited contexts such as Bangladesh, thereby democratizing advanced sign language recognition technology.

6.3 Recommendations for Future Work

This research has successfully built a strong word-level Bangla Sign Language (BdSL) recognition model with a mid-fusion approach of 3D CNNs and TCNs. There are still many ways in which the research can be extended in order to improve its real-world usability, as well as its scalability and usability by the differently abled.

Transition to Continuous Sign Language Recognition The current system operates at the word level, recognizing isolated signs. However, natural communication involves continuous sentence structures where signs flow into one another, introducing co-articulation effects (where the end of one sign influences the start of

the next). Future work should focus on developing Sequence-to-Sequence (Seq2Seq) models or Transformer-based architectures capable of **Continuous Sign Language Recognition (CSLR)**. This would involve translating full sentences rather than discrete words, requiring the integration of language models to handle grammar and syntax correction for Bangla.
Currently, the model is limited to word-level recognition of isolated signs. In reality, signers use long sequences of signs that flow together to form sentences, creating "co-articulation" (in which the last part of a sign affects how the first part of the following sign will appear). Thus, it is recommended that future researchers develop Seq2Seq models or Transformer-based architectures to support **Continuous Sign Language Recognition (CSLR)**. This will translate entire sentences instead of individual words, and will require integrating language models to provide both grammatical and syntactic correction for Bangla.

Data Size Expansion & Data Diversity While the BdSL-300 dataset has made a major contribution, it is still much smaller than other large datasets such as Kinetics-400 or WLASL. A fully comprehensive translation system will require an expansion of the vocabulary size to reach several thousand words, as well as to capture all types of specialized terminology (for example, medical, legal, or technical terminology). In addition to expanding the vocabulary size, it is also important to increase the variety of the data set. Future data collections should include:

- **A Larger Group of Signers:** To capture a larger group of signing speed and style.
- **Sign Language Variations from Different Areas of Bangladesh:** To account for variation in signs from the different areas of Bangladesh.
- **Signers who are Left Handed:** To ensure that the model can generalize well to left handed signers without needing to mirror their hands manually.

Optimization for Edge and Mobile Devices Accessibility is one of the main goals of this project. The current model operates with good efficiency in high-end GPU environments. However, in order to be deployed on inexpensive smartphones or IoT devices (such as Raspberry Pi), further optimizations will need to be conducted by using model compression techniques such as:

- **Quantization:** Using reduced-precision model weights (e.g., from FP32 to INT8) to reduce the amount of memory needed.
- **Network Pruning:** Eliminating unnecessary connections within the neural network to increase inference speeds.
- **Knowledge Distillation:** Training lightweight 'student' models to mimic the performance of the larger 'teacher' model.

Robustness to Unpredictable Real World Variables Deployment in the real world involves many environmental factors that are not present in controlled data sets used in training. Future iterations of the system must increase its robustness to:

- **Occlusions:** The ability to accurately interpret hand actions when they partially block the face or cross over each other.
- **Variable Lighting:** The ability to maintain accuracy in both low light and overly bright lighting conditions that occur during normal day-to-day operation.
- **Variable Background:** Increasing the ability of the segmentation pipeline to process complex and/or changing background images without slowing down the inference speed of the application.

Adaptive Fine-Tuning and Personalization Additionally, incorporating mechanisms of few-shot learning or adaptive fine-tuning would allow the system to learn from the specific user over time. This would provide the opportunity for the system to adapt to a user's unique signing style and/or physical limitations, ultimately providing an improved user experience.

Future research that addresses these issues will provide a path to transform academic prototypes into effective and usable communication systems for the Bengali Deaf community.

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